Official College Board Digital SAT Question Bank — Released Questions

Predicted for the August SAT — CLC Educational Institute

Geometry and Trigonometry

Math | 51 questions with full Rule / Tip / Trap / POE / Desmos explanations

The CLC Method

Every question in this file is broken down the same way, so you build one consistent habit instead of guessing question to question:

The Rule — the underlying math principle or formula being tested. Learn the rule, not just this one answer, and you'll be able to answer every future question of this type.

The Tip — a concrete move you can make before diving into algebra, so you solve efficiently instead of the long way.

Step-by-Step Solution — every single algebra step written out in full, with nothing skipped — so you can follow exactly how the answer was reached, not just see the final number.

The Trap — the single wrong answer most students actually pick, and the specific arithmetic or logical slip that produces it.

POE (multiple-choice questions only) — a full walkthrough of the other wrong choices, showing exactly what error produces each one. Some Math questions are Student-Produced Response ('grid-in') questions with no answer choices at all — for those, there's no POE section, just the full solution.

Solving with Desmos — exact, specific instructions for using the built-in Desmos graphing calculator (available on every Digital SAT Math question) to solve or verify the question — including the exact expressions to type in, not just a general suggestion to 'graph it.'

Area and volume

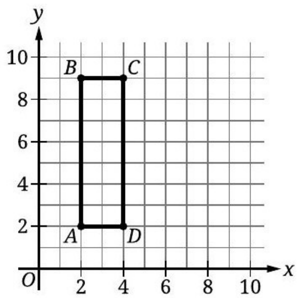
16 questions · Easy 5 / Medium 3 / Hard 8

Area and volume — Q1 [Easy] (ID: 124bc42b)

Question ID: 124bc42b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Easy

Question



Rectangle $ABCD$ shown has a length of 7 units and a width of 2 units. What is the area, in square units, of rectangle $ABCD$?

Answer

A. 5

B. 9

C. 14

D. 28

Correct Answer: C

The Rule: The area of a rectangle is always calculated by multiplying its length by its width: Area = length × width. This formula works regardless of where the rectangle is positioned on a coordinate grid.

The Tip: When a rectangle is shown on a coordinate grid, you don't need to use the coordinates at all if the length and width are already given directly in the problem text — the picture is just there to help you visualize the shape, not to require you to calculate distances from coordinates.

Why C is right: The problem directly states the rectangle has a length of 7 units and a width of 2 units. Using the area formula: Area = length × width = 7 × 2 = 14.

Step-by-Step Solution:

Step 1: Identify the given dimensions: length = 7 units, width = 2 units.

Step 2: Apply the area formula for a rectangle: Area = length × width.

Step 3: Substitute the values: Area = 7 × 2 = 14.

Step 4: This matches choice C.

The Trap: Choice D (28) is the most commonly picked wrong answer — it typically comes from mistakenly using the PERIMETER formula (2 × length + 2 × width = 2×7 + 2×2 = 18, not quite 28 either) or from doubling the correct area by mistake, rather than the simple area formula.

POE — Eliminate the other three:

✗ A — 5 doesn't correspond to any calculation using the given dimensions — likely confused with the difference between length and width (7−2=5) rather than their product.

✗ **B —** 9 comes from ADDING the length and width (7+2=9) instead of multiplying them — this would be relevant for perimeter-related calculations, not area.

✗ **D —** 28 doesn't match a direct area calculation — possibly from doubling the correct area (14×2=28) or another multiplication error.

Solving with Desmos:

Step 1: Open Desmos and calculate the area directly: type 7*2

Step 2: Press Enter — Desmos will evaluate this to 14, confirming the area and matching choice C.

Step 3: If you want to verify using the coordinate points shown (A, B, C, D), you can also calculate the distance between two adjacent points directly in Desmos using the distance formula, though since the length and width are given directly in the problem, this extra step isn't necessary here.

Area and volume — Q2 [Easy] (ID: c8ab85ad)

Question ID: c8ab85ad

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Easy

Question

Which expression represents the area, in square inches, of a square with a side length of 35 inches?

Answer

A. $\sqrt{35}$

B. $35 + 35$

C. $(4)(35)$

D. 35^2

Correct Answer: D

The Rule: The area of a square is always calculated by squaring the side length: Area = $(\text{side length})^2$. This is a special case of the rectangle area formula, since a square is a rectangle where length and width are equal.

The Tip: For 'which expression represents' questions, you don't need to calculate a final numeric answer — just identify the correct FORMULA setup and match it to the choices. Here, recognizing 'area of a square = side^2' immediately points you to the answer without any arithmetic.

Why D is right: For a square with side length 35 inches, the area formula gives Area = 35^2, which is exactly what choice D shows.

Step-by-Step Solution:

Step 1: Recall the area formula for a square: Area = $(\text{side length})^2$.

Step 2: Substitute the given side length: Area = 35^2.

Step 3: This matches choice D exactly — no further calculation is needed since the question asks for the EXPRESSION, not a numeric value.

The Trap: Choice C, (4)(35), is the most commonly picked wrong answer — it correctly uses the number 4 (associated with a square's four equal sides) but incorrectly applies it as a multiplication factor, which would actually calculate the PERIMETER of a square with side 35, not its area.

POE — Eliminate the other three:

✗ **A —** $\sqrt{35}$ represents a square ROOT operation, which would be used to find a side length FROM a given area — but here we're going the opposite direction (finding area FROM a given side length), so this is backwards.

✗ **B —** 35 + 35 represents adding two sides together, which doesn't correspond to any standard area or perimeter formula for a square in this simple form.

✗ **C —** (4)(35) represents multiplying the side length by 4 — this is actually the PERIMETER formula for a square (4 × side), not the area formula.

Solving with Desmos:

Step 1: This question asks for an expression, not a numeric value, so Desmos verification is optional — but you can still check that your chosen expression gives a sensible area. Open Desmos and type: 35^2

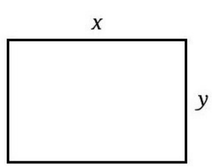
Step 2: Press Enter — Desmos evaluates this to 1225, which is the actual numeric area in square inches, confirming that 35^2 (choice D) is the correct way to represent this square's area.

Area and volume — Q3 [Easy] (ID: f3f06c3a)

Question ID: f3f06c3a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Easy

Question



Note: Figure not drawn to scale.

In the rectangle shown, $x = 7$ centimeters and $y = 5$ centimeters. What is the area, in square centimeters, of the rectangle?

Answer

A. 7

B. 12

C. 35

D. 40

Correct Answer: C

The Rule: The area of a rectangle is always calculated by multiplying its length by its width: Area = length × width. This holds true regardless of how the rectangle is labeled or oriented in a figure.

The Tip: When a figure labels its dimensions with variables (like x and y here) and then gives you specific numeric values for those variables in the text, simply substitute the numbers directly into the area formula — don't worry about matching the exact pixel proportions shown in the (often not-to-scale) figure.

Why C is right: The rectangle's dimensions are given as x = 7 centimeters and y = 5 centimeters. Using the area formula: Area = x × y = 7 × 5 = 35.

Step-by-Step Solution:

Step 1: Identify the given dimensions: x = 7 centimeters, y = 5 centimeters.

Step 2: Apply the area formula for a rectangle: Area = length × width = x × y.

Step 3: Substitute the values: Area = 7 × 5 = 35.

Step 4: This matches choice C.

The Trap: Choice B (12) is the most commonly picked wrong answer — it results from ADDING the two dimensions (7 + 5 = 12) instead of multiplying them, confusing the area formula with the process for finding half the perimeter.

POE — Eliminate the other three:

✗ **A —** 7 is simply the value of x alone — this ignores the y dimension entirely.

✗ **B —** 12 comes from adding x and y (7+5=12) instead of multiplying them — this is a perimeter-related calculation, not area.

✗ **D —** 40 doesn't correspond to a direct calculation using x=7 and y=5 — possibly confused with different dimension values.

Solving with Desmos:

Step 1: Open Desmos and calculate the area directly: type 7*5

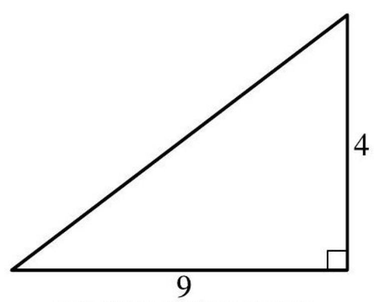
Step 2: Press Enter — Desmos will evaluate this to 35, confirming the area and matching choice C.

Area and volume — Q4 **[Easy]** (ID: cab8f907)

Question ID: cab8f907

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Easy

Question



Note: Figure not drawn to scale.

The figure shows the lengths, in units, of two sides of a triangle. What is the area, in square units, of the triangle?

Answer

A. 18

B. 13

C. 9

D. 4

Correct Answer: A

The Rule: The area of a triangle is always calculated using the formula: Area = (1/2) × base × height, where the base and height must be PERPENDICULAR to each other (meeting at a right angle, as shown by the small square symbol in the figure).

The Tip: Look carefully for the right-angle marker (small square) in a triangle figure — it tells you exactly which two sides are the 'base' and 'height' to use in the area formula. Not every pair of given side lengths in a triangle problem are automatically the base and height; they must be the two sides meeting at that right angle.

Why A is right: The right-angle marker in the figure shows that the sides of length 9 and 4 meet at a right angle, making them the base and height. Using the formula: Area = (1/2) × 9 × 4 = (1/2) × 36 = 18.

Step-by-Step Solution:

Step 1: Identify the base and height from the figure: the right-angle marker confirms the sides labeled 9 and 4 are perpendicular to each other, making them the base and height.

Step 2: Apply the triangle area formula: Area = (1/2) × base × height.

Step 3: Substitute the values: Area = (1/2) × 9 × 4.

Step 4: Calculate: 9 × 4 = 36, then (1/2) × 36 = 18.

Step 5: This matches choice A.

The Trap: Choice B (13) is the most commonly picked wrong answer — it results from simply ADDING the two given side lengths (9 + 4 = 13) instead of correctly applying the triangle area formula, which requires multiplying and then halving.

POE — Eliminate the other three:

✗ B — 13 comes from adding the two given lengths (9+4=13) — this doesn't correspond to any part of the correct area formula.

✗ C — 9 is simply the base length alone, without multiplying by the height or applying the 1/2 factor.

✗ D — 4 is simply the height alone, without multiplying by the base or applying the 1/2 factor.

Solving with Desmos:

Step 1: Open Desmos and calculate the area directly using the triangle area formula: type (1/2)*9*4

Step 2: Press Enter — Desmos will evaluate this to 18, confirming the area and matching choice A.

Area and volume — Q5 [Easy] (ID: 17049054)

Question ID: 17049054

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Easy

Question

The length of a side of square X is 9 centimeters. The area of rectangle Y is 32 square centimeters. What is the total area, in square centimeters, of square X and rectangle Y?

Answer

A. 145

B. 113

C. 82

D. 81

Correct Answer: B

The Rule: To find the TOTAL area of two separate shapes, first calculate each shape's area individually using its own appropriate formula, then ADD the two areas together.

The Tip: When a problem describes two different shapes (like a square and a separately-defined rectangle), resist the urge to combine their given measurements before calculating — find each area completely on its own first, then add the two final results together.

Why B is right: The area of square X is (side length)^2 = 9^2 = 81 square centimeters. The area of rectangle Y is given directly as 32 square centimeters. The total combined area is 81 + 32 = 113.

Step-by-Step Solution:

Step 1: Calculate the area of square X using the given side length: Area = side^2 = 9^2 = 81 square centimeters.

Step 2: Note the area of rectangle Y, which is given directly in the problem: 32 square centimeters.

Step 3: Add the two areas together to find the total: 81 + 32 = 113.

Step 4: This matches choice B.

The Trap: Choice D (81) is the most commonly picked wrong answer — it's the area of square X ALONE, calculated correctly but then the student forgets to add rectangle Y's area (32) to get the actual TOTAL area the question asks for.

POE — Eliminate the other three:

✗ **A —** 145 doesn't correspond to a correct combination of 81 and 32 — likely from a different, incorrect calculation of the square's area before adding.

✗ **C —** 82 is very close to the square's correct area (81) but off by one — possibly from a minor arithmetic slip, and still missing rectangle Y's area entirely.

✗ **D —** 81 is the area of square X ALONE ($9^2 = 81$) — this is a correct partial calculation, but it forgets to ADD rectangle Y's area (32) to find the TOTAL area the question asks for.

Solving with Desmos:

Step 1: Open Desmos and calculate the square's area first: type 9^2

Step 2: Press Enter — this confirms 81 square centimeters for square X.

Step 3: Add rectangle Y's given area: type 81+32

Step 4: Press Enter — Desmos will evaluate this to 113, confirming the total combined area and matching choice B.

Area and volume — Q6 [Medium] [Grid-In] (ID: 25cc5327)

Question ID: 25cc5327

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

The base of a right rectangular pyramid has a length of 27 centimeters and a width of 2 centimeters. The height of this pyramid is 7 centimeters. What is the volume, in cubic centimeters, of this pyramid?

Correct Answer: 126 (Student-Produced Response — no answer choices given)

The Rule: The volume of any pyramid is calculated using the formula: Volume = (1/3) × (base area) × height. For a RECTANGULAR pyramid, the base area is simply length × width.

The Tip: Always calculate the base area FIRST as its own separate step, then plug that single number into the pyramid volume formula — trying to do everything in one big calculation increases the chance of a multiplication error.

Why this is the answer: The rectangular base has area = length × width = 27 × 2 = 54 square centimeters. Using the pyramid volume formula: Volume = (1/3) × 54 × 7 = (1/3) × 378 = 126.

Step-by-Step Solution:

Step 1: Calculate the base area of the rectangular pyramid: base area = length × width = 27 × 2 = 54 square centimeters.

Step 2: Apply the pyramid volume formula: Volume = (1/3) × base area × height.

Step 3: Substitute the values: Volume = (1/3) × 54 × 7.

Step 4: Calculate 54 × 7 = 378, then (1/3) × 378 = 126.

Step 5: This is a grid-in question — enter 126 as your final answer.

The Trap: A common error on this problem is forgetting the (1/3) factor entirely, calculating just base area × height (54 × 7 = 378) as if it were a rectangular PRISM instead of a pyramid — pyramids always have exactly one-third the volume of a prism with the same base and height.

Solving with Desmos:

Step 1: Open Desmos and calculate the base area first: type 27*2 — this confirms 54 square centimeters.

Step 2: Calculate the full volume using the pyramid formula: type (1/3)*54*7

Step 3: Press Enter — Desmos will evaluate this to 126, confirming your grid-in answer.

Area and volume — Q7 [Medium] (ID: 42155bac)

Question ID: 42155bac

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

The area of one of the bases of a right circular cylinder is 81 square centimeters, and the volume of the cylinder is $2{,}187$ cubic centimeters. What is the height, in centimeters, of the cylinder?

Answer

A. 27

B. $2{,}025$

C. $2{,}106$

D. $177{,}147$

Correct Answer: A

The Rule: The volume of a cylinder is calculated using the formula: Volume = (base area) × height. If you're given the volume and the base area, you can find the height by dividing: height = Volume ÷ base area.

The Tip: Recognize that a cylinder's volume formula is structurally the same as a rectangular prism's (base area × height) — this makes it easy to rearrange and solve for height using simple division once you have the volume and base area.

Why A is right: Using the formula Volume = base area × height, and solving for height: height = Volume ÷ base area = 2,187 ÷ 81 = 27.

Step-by-Step Solution:

Step 1: Recall the cylinder volume formula: Volume = base area × height.

Step 2: Rearrange this formula to solve for height: height = Volume ÷ base area.

Step 3: Substitute the given values: height = 2,187 ÷ 81.

Step 4: Calculate: 2,187 ÷ 81 = 27.

Step 5: This matches choice A.

The Trap: Choice B (2,025) is the most commonly picked wrong answer — it likely results from SUBTRACTING (2,187 − 81 = 2,106, close to choice C actually) or another incorrect operation instead of correctly DIVIDING the volume by the base area.

POE — Eliminate the other three:

✗ B — 2,025 doesn't correspond to a correct division of 2,187 by 81 — likely from an unrelated arithmetic error or confusing which operation to use.

✗ C — 2,106 comes from SUBTRACTING the base area from the volume (2,187 − 81 = 2,106) instead of correctly dividing to solve for height.

✗ D — 177,147 comes from MULTIPLYING the volume by the base area (2,187 × 81) instead of dividing — the exact opposite, incorrect operation.

Solving with Desmos:

Step 1: Open Desmos and calculate the height directly: type 2187/81

Step 2: Press Enter — Desmos will evaluate this to 27, confirming the height and matching choice A.

Area and volume — Q8 [Medium] [Grid-In] (ID: 1b687ae3)

Question ID: 1b687ae3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

The area of a triangle is 36 square units. If the height of this triangle is 2 units, what is the length, in units, of the base of this triangle?

Correct Answer: 36 (Student-Produced Response — no answer choices given)

The Rule: The area of a triangle is always calculated using the formula: Area = (1/2) × base × height. If you're given the area and the height, you can find the base by rearranging: base = (2 × Area) ÷ height.

The Tip: When solving backwards from a given area to find a missing base or height, always multiply the area by 2 FIRST to cancel out the (1/2) in the formula, then divide by whichever dimension you already know.

Why this is the answer: Using the triangle area formula Area = (1/2) × base × height, and solving for base: base = (2 × Area) ÷ height = (2 × 36) ÷ 2 = 72 ÷ 2 = 36.

Step-by-Step Solution:

Step 1: Recall the triangle area formula: Area = (1/2) × base × height.

Step 2: Rearrange this formula to solve for the base: base = (2 × Area) ÷ height.

Step 3: Substitute the given values: base = (2 × 36) ÷ 2.

Step 4: Calculate the numerator first: 2 × 36 = 72.

Step 5: Divide: 72 ÷ 2 = 36.

Step 6: This is a grid-in question — enter 36 as your final answer.

The Trap: A common error on this problem is forgetting to multiply the area by 2 before dividing by the height, which would incorrectly give base = 36 ÷ 2 = 18 instead of the correct 36 — always remember the factor of 2 needed to 'undo' the (1/2) in the area formula.

Solving with Desmos:

Step 1: Open Desmos and calculate the base directly using the rearranged formula: type (2*36)/2

Step 2: Press Enter — Desmos will evaluate this to 36, confirming your grid-in answer.

Area and volume — Q9 [Hard] (ID: 1fe41c6b)

Question ID: 1fe41c6b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

The table gives the areas and perimeters of two similar rectangles, where n is a constant.

	Area (square inches)	Perimeter (inches)
Rectangle A	630	210
Rectangle B	$2{,}520$	n

What is the value of n?

Answer

A. 2,100

B. 1,680

C. 840

D. 420

Correct Answer: D

The Rule: For SIMILAR figures, the ratio of their PERIMETERS equals the ratio of their corresponding LINEAR dimensions (the 'scale factor'), while the ratio of their AREAS equals the SQUARE of that same scale factor. Always find the scale factor from the area ratio FIRST (by taking a square root), then apply that same scale factor directly to the perimeter.

The Tip: When given both area and perimeter information for similar figures, use the AREA ratio to find the scale factor (by taking its square root), since area ratios reveal the TRUE linear scale factor more reliably than trying to reason from perimeter alone.

Why D is right: The area ratio between Rectangle B and Rectangle A is 2,520 ÷ 630 = 4. Since area ratio equals the SQUARE of the scale factor, the actual linear scale factor is $\sqrt{4} = 2$. Since perimeter scales linearly (not squared), Rectangle B's perimeter is 2 times Rectangle A's perimeter: $n = 210 \times 2 = 420$.

Step-by-Step Solution:

Step 1: Calculate the area ratio between the two similar rectangles: Rectangle B's area ÷ Rectangle A's area = 2,520 ÷ 630 = 4.

Step 2: Since area scales as the SQUARE of the linear scale factor, find the actual scale factor by taking the square root of the area ratio: $\sqrt{4} = 2$.

Step 3: Since perimeter scales LINEARLY (directly proportional to the scale factor, not squared), multiply Rectangle A's perimeter by this same scale factor: $n = 210 \times 2$.

Step 4: Calculate: $210 \times 2 = 420$.

Step 5: This matches choice D.

The Trap: Choice A (2,100) is the most commonly picked wrong answer — it results from directly multiplying Rectangle A's perimeter by the AREA ratio (4) instead of correctly using the LINEAR scale factor (2) — perimeter should only be multiplied by the linear scale factor, not the area ratio.

POE — Eliminate the other three:

✗ **A** — 2,100 comes from multiplying 210 by the area ratio (4) directly — but perimeter scales with the LINEAR scale factor (2), not the area ratio itself.

✗ **B** — 1,680 doesn't correspond to a clean derivation from this problem — possibly from an unrelated calculation error.

✗ **C** — 840 comes from multiplying 210 by 4 and then dividing by something, or another miscalculation that doesn't correctly apply the square-root relationship between area ratio and linear scale factor.

Solving with Desmos:

Step 1: Open Desmos and calculate the area ratio first: type 2520/630 — this confirms the ratio is 4.

Step 2: Find the linear scale factor by taking the square root: type sqrt(4) — this confirms the scale factor is 2.

Step 3: Apply this scale factor to Rectangle A's perimeter: type 210*2

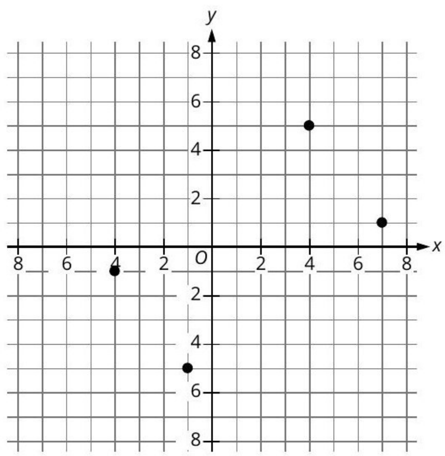
Step 4: Press Enter — Desmos will evaluate this to 420, confirming n = 420 and matching choice D.

Area and volume — Q10 [Hard] [Grid-In] (ID: 30aa51b7)

Question ID: 30aa51b7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question



A rectangle is formed by the four points shown. What is the area, in square units, of the rectangle?

Correct Answer: 50 (Student-Produced Response — no answer choices given)

The Rule: When a rectangle's vertices are given as coordinate points (not aligned with the x and y axes), find the side lengths using the DISTANCE FORMULA between adjacent vertices, then multiply those two side lengths together for the area — just like any other rectangle.

The Tip: For a 'tilted' rectangle on a coordinate grid, identify which pairs of points are adjacent (connected by a side) versus diagonal (opposite corners) — adjacent sides should be PERPENDICULAR to each other, which you can verify by checking that their slopes multiply to −1.

Why this is the answer: Reading the four points from the graph: approximately (−4,−1), (4,5), (7,1), and (−1,−5). Using the distance formula, one side has length $\sqrt{[(4-(-4))^2 + (5-(-1))^2]} = \sqrt{(64+36)} = \sqrt{100} = 10$, and the adjacent side has length $\sqrt{[(7-4)^2 + (1-5)^2]} = \sqrt{(9+16)} = \sqrt{25} = 5$. The area is 10 × 5 = 50.

Step-by-Step Solution:

Step 1: Read the four vertex coordinates from the graph: approximately (−4,−1), (4,5), (7,1), and (−1,−5).

Step 2: Calculate the length of one side using the distance formula between two adjacent vertices, (−4,−1) and (4,5): distance = $\sqrt{[(4-(-4))^2 + (5-(-1))^2]} = \sqrt{[(8)^2 + (6)^2]} = \sqrt{(64+36)} = \sqrt{100} = 10$.

Step 3: Calculate the length of the adjacent side between (4,5) and (7,1): distance = $\sqrt{[(7-4)^2 + (1-5)^2]} = \sqrt{[(3)^2 + (-4)^2]} = \sqrt{(9+16)} = \sqrt{25} = 5$.

Step 4: Verify these two sides are perpendicular (confirming it's truly a rectangle) by checking their slopes multiply to −1: slope of first side = 6/8 = 3/4; slope of second side = −4/3. Product: (3/4)×(−4/3) = −1. Confirmed perpendicular.

Step 5: Calculate the area by multiplying the two side lengths: Area = 10 × 5 = 50.

Step 6: This is a grid-in question — enter 50 as your final answer.

The Trap: A common error on this problem is misreading the exact coordinates of the points from the graph (since they're not always sitting exactly on grid intersections), which throws off the entire distance calculation — always double check each point's position carefully against the grid lines.

Solving with Desmos:

Step 1: Open Desmos and plot the four points you read from the graph: (-4,-1), (4,5), (7,1), (-1,-5)

Step 2: To calculate the distance between two adjacent points, type: sqrt((4-(-4))^2+(5-(-1))^2) — this confirms one side length is 10.

Step 3: Type: sqrt((7-4)^2+(1-5)^2) — this confirms the adjacent side length is 5.

Step 4: Calculate the area: type 10*5 — Desmos will evaluate this to 50, confirming your grid-in answer.

Area and volume — Q11 [Hard] [Grid-In] (ID: 179edb12)

Question ID: 179edb12

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A right circular cone has a height of 22 centimeters (cm) and a base with a diameter of $18\ \text{cm}$. The volume of this cone is $n\pi\ \text{cm}^3$. What is the value of n?

Correct Answer: 594 (Student-Produced Response — no answer choices given)

The Rule: The volume of a cone is calculated using the formula: Volume = (1/3) × π × r² × height, where r is the RADIUS of the circular base (not the diameter). Always remember to divide the given diameter by 2 to get the radius before plugging into the formula.

The Tip: When a cone problem gives you the DIAMETER instead of the radius, immediately divide by 2 as your very first step, before touching the volume formula — using the diameter directly in place of the radius is one of the most common errors on cone and cylinder problems.

Why this is the answer: The diameter is 18 cm, so the radius is 18 ÷ 2 = 9 cm. Using the cone volume formula: Volume = (1/3) × π × 9² × 22 = (1/3) × π × 81 × 22 = (1/3) × 1,782 × π = 594π. Since the volume is given as nπ, this means n = 594.

Step-by-Step Solution:

Step 1: Convert the given diameter to radius: radius = diameter ÷ 2 = 18 ÷ 2 = 9 cm.

Step 2: Recall the cone volume formula: Volume = (1/3) × π × r² × height.

Step 3: Substitute the values: Volume = (1/3) × π × 9² × 22.

Step 4: Calculate 9² = 81.

Step 5: Multiply: 81 × 22 = 1,782.

Step 6: Multiply by (1/3): (1/3) × 1,782 = 594.

Step 7: The volume is 594π cubic centimeters, so comparing to the given form nπ, we get n = 594.

Step 8: This is a grid-in question — enter 594 as your final answer.

The Trap: A common error on this problem is using the given diameter (18) directly as the radius in the formula, instead of first dividing by 2 to get the correct radius (9) — this would lead to an answer four times too large, since the radius is squared in the formula.

Solving with Desmos:

Step 1: Open Desmos and calculate the radius first: type 18/2 — this confirms the radius is 9 cm.

Step 2: Calculate the volume coefficient (the value of n): type (1/3)*9^2*22

Step 3: Press Enter — Desmos will evaluate this to 594, confirming n = 594 for your grid-in answer. (Desmos will show 594 as a plain number since you're calculating the coefficient of π, not the full numeric volume.)

Area and volume — Q12 [Hard] (ID: e6c557f9)

Question ID: e6c557f9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

The edge length, in inches, of cube Y is $\frac{3}{88}$ the edge length, in inches, of cube X. The surface area, in square inches, of cube Y is n times the surface area, in square inches, of cube X. What is the value of n?

Answer

A. $\frac{9}{7,744}$

B. $\frac{27}{3,872}$

C. $\frac{3}{88}$

D. $\frac{9}{44}$

Correct Answer: A

The Rule: For SIMILAR three-dimensional shapes, the ratio of their SURFACE AREAS equals the SQUARE of the ratio of their corresponding linear dimensions (like edge length) — this is the 3D analog of how area scales for similar 2D shapes.

The Tip: Whenever you see a ratio comparing similar shapes' surface areas (a 2D measurement), square whatever linear (1D) ratio you're given — don't use the linear ratio directly, and don't cube it either (cubing is for VOLUME ratios, not surface area).

Why A is right: The edge length ratio of cube Y to cube X is 3/88. Since surface area scales as the SQUARE of the linear ratio, the surface area ratio $n = (3/88)^2 = 9/7,744$.

Step-by-Step Solution:

Step 1: Identify the given linear (edge length) ratio: cube Y's edge is 3/88 of cube X's edge.

Step 2: Recall that surface area scales as the SQUARE of the linear scale factor for similar 3D shapes.

Step 3: Square the given ratio: $n = (3/88)^2 = (3^2)/(88^2) = 9/7,744$.

Step 4: This matches choice A.

The Trap: Choice C (3/88) is the most commonly picked wrong answer — it uses the LINEAR edge-length ratio directly as the answer, without squaring it to correctly account for the fact that surface area is a two-dimensional measurement.

POE — Eliminate the other three:

✗ **B** — 27/3,872 comes from CUBING the ratio ($3^3/88^2$ or a similar mismatched operation) instead of correctly squaring both the numerator and denominator — cubing would be appropriate for VOLUME ratios, not surface area.

✗ **C** — 3/88 is simply the original linear edge-length ratio, used directly without squaring — this doesn't account for surface area being a two-dimensional measurement.

✗ **D** — 9/44 doesn't correctly square the denominator ($88^2 = 7,744$, not 44) — this likely results from only partially squaring the ratio.

Solving with Desmos:

Step 1: Open Desmos and calculate the surface area ratio by squaring the given edge-length ratio: type (3/88)^2

Step 2: Press Enter — Desmos will evaluate this to a decimal (approximately 0.000388...), and you can compare it to 9/7744 by typing that as well to confirm they match exactly, verifying choice A.

Area and volume — Q13 [Hard] [Grid-In] (ID: 73e1b793)

Question ID: 73e1b793

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

Right rectangular prism P is similar to right rectangular prism Q. The volume of prism P is $972 \text{ cubic centimeters } (\text{cm}^3)$, and the volume of prism Q is 36 cm^3. One edge of prism P has a length of 12 cm. What is the length, in cm, of the corresponding edge of prism Q?

Correct Answer: 4 (Student-Produced Response — no answer choices given)

The Rule: For SIMILAR three-dimensional shapes, the ratio of their VOLUMES equals the CUBE of the ratio of their corresponding linear dimensions — this is different from area, which uses the SQUARE of the ratio.

The Tip: When given a volume ratio for similar solids and asked to find a linear dimension, take the CUBE ROOT of the volume ratio to find the actual linear scale factor — don't take the square root (that's for area problems) or use the volume ratio directly.

Why this is the answer: The volume ratio of prism P to prism Q is 972/36 = 27. Since volume scales as the CUBE of the linear scale factor, the actual linear scale factor is the cube root of 27, which is 3. This means every edge of prism P is 3 times the corresponding edge of prism Q. Since one edge of P is 12 cm, the corresponding edge of Q is 12 ÷ 3 = 4 cm.

Step-by-Step Solution:

Step 1: Calculate the volume ratio between the two similar prisms: prism P's volume ÷ prism Q's volume = 972 ÷ 36 = 27.

Step 2: Since volume scales as the CUBE of the linear scale factor, find the actual scale factor by taking the cube root of the volume ratio: the cube root of 27 is 3 (since 3×3×3 = 27).

Step 3: This means every linear dimension of prism P is 3 times the corresponding dimension of prism Q.

Step 4: Since the given edge of prism P is 12 cm, find the corresponding edge of prism Q by dividing by the scale factor: 12 ÷ 3 = 4.

Step 5: This is a grid-in question — enter 4 as your final answer.

The Trap: A common error on this problem is taking the SQUARE ROOT of the volume ratio (√27 ≈ 5.2) instead of the correct CUBE ROOT (3) — remembering that volume is a three-dimensional measurement (requiring a cube root to find the linear scale) is the key insight here.

Solving with Desmos:

Step 1: Open Desmos and calculate the volume ratio first: type 972/36 — this confirms the ratio is 27.

Step 2: Find the linear scale factor by taking the cube root: type 27^(1/3) — Desmos will show this equals 3.

Step 3: Calculate the corresponding edge of prism Q: type 12/3

Step 4: Press Enter — Desmos will evaluate this to 4, confirming your grid-in answer.

Area and volume — Q14 [Hard] (ID: 9fdc4eaa)

Question ID: 9fdc4eaa

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A right square pyramid has a total surface area of $70,560$ square inches, and the combined surface area of the four lateral faces of this pyramid is $38,160$ square inches. What is the height, in inches, of this pyramid?

Answer

A. 56

B. 90

C. 106

D. 180

Correct Answer: A

The Rule: For a right square pyramid, the total surface area equals the base area PLUS the combined lateral (side) face area. The slant height relates to the pyramid's actual height through the Pythagorean theorem, using half the base's side length as the other leg of a right triangle.

The Tip: Break this multi-step problem into clear stages: (1) find the base area by subtracting lateral area from total surface area, (2) find the base's side length by taking the square root, (3) find the slant height using the lateral area formula, and (4) use the Pythagorean theorem to find the actual height. Tackling one stage at a time prevents getting overwhelmed.

Why A is right: The base area is $70,560 - 38,160 = 32,400$ square inches, so the base's side length is $\sqrt{32,400} = 180$ inches. The lateral area formula for a square pyramid is $2 \times$ (side length) $\times$ (slant height) $= 38,160$, so slant height $= 38,160 \div (2\times180) = 38,160 \div 360 = 106$. Using the Pythagorean theorem with half the base (90) and the slant height (106): $\text{height}^2 = 106^2 - 90^2 = 11,236 - 8,100 = 3,136$, so height $= \sqrt{3,136} = 56$.

Step-by-Step Solution:

Step 1: Find the base area by subtracting the lateral surface area from the total surface area: base area $= 70,560 - 38,160 = 32,400$ square inches.

Step 2: Since the base is a square, find its side length by taking the square root: side length $= \sqrt{32,400} = 180$ inches.

Step 3: Use the lateral surface area formula for a square pyramid (4 triangular faces, each with area $(1/2) \times$ base side $\times$ slant height, combined $= 2 \times$ base side $\times$ slant height): $38,160 = 2 \times 180 \times$ slant height.

Step 4: Solve for slant height: slant height $= 38,160 \div 360 = 106$ inches.

Step 5: Set up the Pythagorean theorem relating the pyramid's height, half the base side length, and the slant height (these three form a right triangle): $\text{height}^2 + (\text{base side}/2)^2 = \text{slant height}^2$.

Step 6: Substitute the values: $\text{height}^2 + 90^2 = 106^2$.

Step 7: Calculate: $\text{height}^2 + 8,100 = 11,236$.

Step 8: Solve for height^2: $\text{height}^2 = 11,236 - 8,100 = 3,136$.

Step 9: Take the square root: height $= \sqrt{3,136} = 56$.

Step 10: This matches choice A.

The Trap: Choice B (90) is the most commonly picked wrong answer — it's actually HALF the base side length (used as an intermediate step in the Pythagorean theorem calculation), not the final height — a student who completes most of the steps correctly but stops early might mistakenly select this intermediate value.

POE — Eliminate the other three:

✗ B — 90 is half the base's side length ($180\div2=90$), an intermediate value used WITHIN the Pythagorean theorem calculation — it's not the final height the question asks for.

✗ C — 106 is the SLANT HEIGHT (a different measurement along the triangular face), not the perpendicular height of the pyramid that the question asks for.

✗ D — 180 is the full side length of the square base — not the height of the pyramid at all.

Solving with Desmos:

Step 1: Open Desmos and calculate the base area: type 70560-38160 — this confirms 32,400 square inches.

Step 2: Find the base's side length: type sqrt(32400) — this confirms 180 inches.

Step 3: Find the slant height: type 38160/(2*180) — this confirms 106 inches.

Step 4: Use the Pythagorean theorem to find the actual height: type sqrt(106^2-90^2)

Step 5: Press Enter — Desmos will evaluate this to 56, confirming the height and matching choice A.

Area and volume — Q15 [Hard] [Grid-In] (ID: f6ca90cc)

Question ID: f6ca90cc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A right prism has a square base with side lengths of 10 inches. The height of the prism is 58 inches. What is the volume, in cubic inches, of the prism?

Correct Answer: 5800 (Student-Produced Response — no answer choices given)

The Rule: The volume of any right prism (including one with a square base) is calculated using the formula: Volume = (base area) × height.

The Tip: For a prism with a square base, calculate the base area first (side length squared) as a separate step, then multiply by the height — breaking the calculation into two clean stages reduces the chance of a multiplication error.

Why this is the answer: The square base has area = side^2 = 10^2 = 100 square inches. Using the prism volume formula: Volume = base area × height = 100 × 58 = 5,800.

Step-by-Step Solution:

Step 1: Calculate the area of the square base: base area = side^2 = 10^2 = 100 square inches.

Step 2: Apply the prism volume formula: Volume = base area × height.

Step 3: Substitute the values: Volume = 100 × 58.

Step 4: Calculate: 100 × 58 = 5,800.

Step 5: This is a grid-in question — enter 5,800 as your final answer.

The Trap: A common error on this problem is using the PERIMETER of the square base (4 × 10 = 40) instead of its AREA (10^2 = 100) when applying the volume formula — always make sure you're using area, not perimeter, for volume calculations.

Solving with Desmos:

Step 1: Open Desmos and calculate the base area first: type 10^2 — this confirms 100 square inches.

Step 2: Calculate the full volume: type 100*58

Step 3: Press Enter — Desmos will evaluate this to 5800, confirming your grid-in answer.

Area and volume — Q16 [Hard] [Grid-In] (ID: 4757123b)

Question ID: 4757123b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

Trapezoid A and trapezoid B are similar. The length of each side of trapezoid A is 41 times the length of the corresponding side of trapezoid B. The area of trapezoid A is how many times as large as the area of trapezoid B?

Correct Answer: 1681 (Student-Produced Response — no answer choices given)

The Rule: For SIMILAR two-dimensional figures (including trapezoids), the ratio of their AREAS equals the SQUARE of the ratio of their corresponding linear dimensions (side lengths).

The Tip: When a problem gives you a linear (side-length) ratio between similar figures and asks about the area relationship, always SQUARE that ratio — don't use it directly, since area is a two-dimensional measurement while side length is only one-dimensional.

Why this is the answer: Since each side of trapezoid A is 41 times the corresponding side of trapezoid B, the linear scale factor is 41. Since area scales as the SQUARE of the linear scale factor, trapezoid A's area is $41^2 = 1{,}681$ times as large as trapezoid B's area.

Step-by-Step Solution:

Step 1: Identify the given linear scale factor: each side of trapezoid A is 41 times the corresponding side of trapezoid B.

Step 2: Recall that area scales as the SQUARE of the linear scale factor for similar figures.

Step 3: Square the given scale factor: $41^2 = 41 \times 41 = 1{,}681$.

Step 4: This means trapezoid A's area is 1,681 times as large as trapezoid B's area.

Step 5: This is a grid-in question — enter 1,681 as your final answer.

The Trap: A common error on this problem is using the linear scale factor (41) directly as the final answer, without squaring it to correctly account for the fact that area is a two-dimensional measurement — remember, side-length ratios must always be squared to find area ratios.

Solving with Desmos:

Step 1: Open Desmos and calculate the area ratio by squaring the given linear scale factor: type 41^2

Step 2: Press Enter — Desmos will evaluate this to 1681, confirming your grid-in answer.

Lines, angles, and triangles

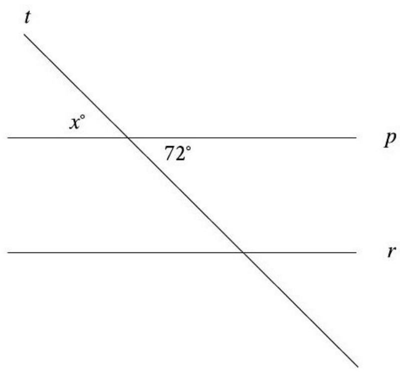
14 questions · Easy 2 / Medium 6 / Hard 6

Lines, angles, and triangles — Q1 [Easy] (ID: 4a141e77)

Question ID: 4a141e77

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure, line p is parallel to line r, and line t intersects both lines. What is the value of x?

Answer

A. 36

B. 72

C. 180

D. 252

Correct Answer: B

The Rule: When two straight lines cross, the two angles directly ACROSS from each other (diagonally opposite, formed by the same two lines) are called vertical angles, and vertical angles are always EQUAL — this is true regardless of whether any other lines in the figure are parallel.

The Tip: Before using any parallel-line facts (corresponding, alternate interior angles), first check if the two labeled angles are simply VERTICAL angles at the SAME intersection point — if so, they're equal by a much more basic rule, and you don't need the parallel-line information at all.

Why B is right: The angles labeled x° and 72° are formed at the same intersection point (where line t crosses line p), positioned diagonally opposite each other — making them vertical angles, which are always equal: x = 72.

Step-by-Step Solution:

Step 1: Identify the two labeled angles' positions: x° and 72° are both located at the single intersection point where line t crosses line p.

Step 2: Determine their relationship: they are diagonally opposite each other at this intersection, making them vertical angles.

Step 3: Apply the vertical angles theorem: vertical angles are always equal, regardless of any other properties in the figure (like whether p is parallel to r).

Step 4: Since 72° is given, x = 72.

Step 5: This matches choice B.

The Trap: Choice A (36) is the most commonly picked wrong answer — it results from incorrectly halving 72°, perhaps confusing this problem with a different angle-bisector scenario that doesn't apply here.

POE — Eliminate the other three:

✗ A — 36 is half of 72 — this might come from mistakenly assuming some bisecting relationship that doesn't apply to vertical angles, which are simply equal, not related by a factor of 2.

✗ **C —** 180 would be the result of treating x and 72 as a LINEAR PAIR (supplementary angles) instead of vertical angles — but based on their diagonal positions in the figure, they are vertical (equal), not supplementary.

✗ **D —** 252 comes from adding 180 + 72 — this doesn't correspond to any standard angle relationship in this figure.

Solving with Desmos:

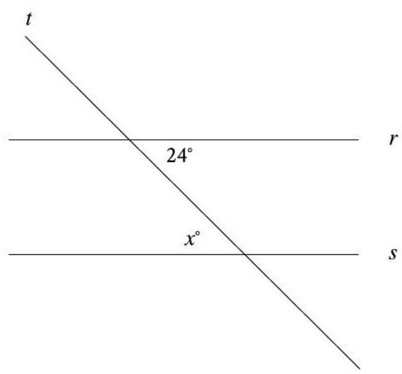
This is a straightforward vertical-angles reading question that doesn't require Desmos calculation — the answer (x=72) follows directly from the vertical angles theorem, visible in the figure itself. If you want to visually confirm, you could sketch the intersection using Desmos's geometry tools by plotting two intersecting lines and observing that the opposite angle pairs are always visually equal.

Lines, angles, and triangles — Q2 [Easy] (ID: d21270da)

Question ID: d21270da

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure, line t intersects lines r and s. Which additional piece of information is sufficient to prove that lines r and s are parallel?

Answer

A. $x < 24$

B. $x = 24$

C. $x > 24$

D. $x - 24 = 360$

Correct Answer: B

The Rule: When a transversal crosses two lines, CORRESPONDING angles (angles in the SAME relative position at each intersection) are equal IF AND ONLY IF the two lines are parallel. This is the Corresponding Angles Postulate, and it works in both directions — equal corresponding angles PROVE the lines are parallel.

The Tip: To identify corresponding angles, imagine 'sliding' one intersection point along the transversal until it lands on top of the other intersection point — whichever two angles end up in the same relative position (same side of the transversal, same side of the crossed lines) are corresponding angles.

Why B is right: The angle 24° (at the r intersection) and the angle x° (at the s intersection) are both positioned in the same relative spot — below their respective horizontal line, on the same side of the transversal t. This makes them corresponding angles. For lines r and s to be parallel, corresponding angles must be equal, so the required condition is x = 24.

Step-by-Step Solution:

Step 1: Identify the positions of the two labeled angles: 24° is at the intersection of t and r; x° is at the intersection of t and s.

Step 2: Compare their relative positions: both angles sit in the same location relative to their own intersection (same side of line t, same side of the horizontal line) — this makes them corresponding angles.

Step 3: Recall the Corresponding Angles Postulate: two lines cut by a transversal are parallel IF AND ONLY IF corresponding angles are equal.

Step 4: Apply this to find the necessary condition: for r and s to be parallel, x must equal 24.

Step 5: This matches choice B.

The Trap: Choice D is the most commonly picked wrong answer — it proposes an equation ($x - 24 = 360$) that doesn't correspond to any standard angle relationship; a difference of 360° doesn't make sense for two angles that should logically be somewhere between 0° and 180°.

POE — Eliminate the other three:

✗ **A —** $x < 24$ doesn't establish a precise EQUALITY needed for the Corresponding Angles Postulate — parallel lines require the angles to be exactly equal, not merely one smaller than the other.

✗ **C —** $x > 24$ has the same problem as choice A — an inequality doesn't provide the precise equality condition needed to prove the lines are parallel.

✗ **D —** $x - 24 = 360$ doesn't correspond to any meaningful angle relationship — angle measures in this context are between 0° and 180°, making a 360° difference nonsensical here.

Solving with Desmos:

This is a conceptual angle-relationship question that doesn't require numerical calculation in Desmos — the answer follows directly from correctly identifying corresponding angles and applying the Corresponding Angles Postulate. If you want to visually verify, you could use Desmos's geometry tools to draw two parallel lines cut by a transversal and confirm that the corresponding angle positions always show equal angle measures.

Lines, angles, and triangles — Q3 [Medium] (ID: e89f63bf)

Question ID: e89f63bf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question

The sum of the measures of two angles in a triangle is $64°$. What is the measure of the third angle?

Answer

A. $26°$

B. $52°$

C. $116°$

D. $128°$

Correct Answer: C

The Rule: The three interior angles of ANY triangle always add up to exactly 180°. If you know the sum of two angles, you can always find the third by subtracting from 180.

The Tip: This is one of the most fundamental and frequently-used facts in geometry — memorize it so thoroughly that you can apply it instantly without hesitation: angle1 + angle2 + angle3 = 180° for every triangle, no exceptions.

Why C is right: Since all three angles of a triangle sum to 180°, and two of the angles sum to 64°, the third angle must be $180 - 64 = 116$.

Step-by-Step Solution:

Step 1: Recall the triangle angle sum theorem: the three interior angles of any triangle always add up to 180°.

Step 2: The problem states two of the angles sum to 64°.

Step 3: Find the third angle by subtracting this sum from 180°: 180 − 64 = 116.

Step 4: This matches choice C.

The Trap: Choice A (26) is the most commonly picked wrong answer — it results from an unrelated calculation (like 90 − 64 = 26, perhaps confusing this with a right-triangle complementary-angle problem) rather than correctly using the 180° triangle angle sum.

POE — Eliminate the other three:

✗ **A —** 26 comes from 90 − 64 = 26 — this incorrectly applies the COMPLEMENTARY angle rule (which only applies to two acute angles in a RIGHT triangle summing to 90°), not the general triangle angle sum of 180° that applies here.

✗ **B —** 52 is exactly half of 104, or possibly half of the correct answer — doesn't correspond to a direct calculation from the given 64° sum.

✗ **D —** 128 is exactly double the correct answer (116×... actually let's check: 128 doesn't have an obvious simple relationship, possibly from a different arithmetic error).

Solving with Desmos:

Step 1: Open Desmos and calculate the third angle directly: type 180-64

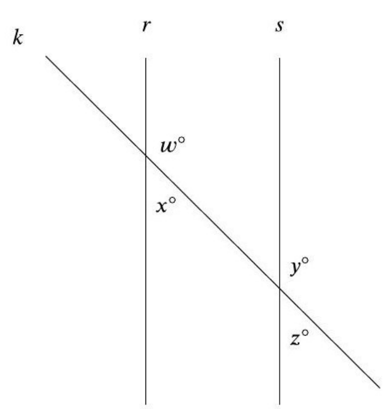
Step 2: Press Enter — Desmos will evaluate this to 116, confirming the third angle and matching choice C.

Lines, angles, and triangles — Q4 **[Medium]** (ID: ecc98c87)

Question ID: ecc98c87

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question



Note: Figure not drawn to scale.

In the figure shown, line k intersects lines r and s. If $w = 147$, which additional piece of information is sufficient to prove that lines r and s are parallel?

Answer

A. $x = 33$

B. $y = 147$

C. $w + y = 180$

D. $y + z = 180$

Correct Answer: B

The Rule: When a transversal crosses two lines, CORRESPONDING angles (same relative position at each intersection) are equal IF AND ONLY IF the two lines are parallel. Identifying which pair of labeled angles are truly 'corresponding' (not vertical, not alternate) is the key step in these problems.

The Tip: Carefully trace the SAME relative position at each intersection point — the angle that occupies the identical corner (upper-left, upper-right, lower-left, or lower-right) relative to the crossing lines at BOTH intersections is the corresponding angle pair, regardless of how far apart the intersections are.

Why B is right: The angle w° (at the r intersection) and the angle y° (at the s intersection) occupy the same relative position at their respective crossings — both in the 'upper-right' region relative to the transversal k and their vertical line. This makes w and y corresponding angles. Since w = 147 (as given), for r and s to be parallel, y must also equal 147.

Step-by-Step Solution:

Step 1: Identify the given angle: w = 147° at the r intersection.

Step 2: Determine which angle at the s intersection occupies the SAME relative position as w — comparing the figure, this is y° (both are in the upper-right region relative to their intersection).

Step 3: Recall the Corresponding Angles Postulate: for two lines cut by a transversal to be parallel, corresponding angles must be equal.

Step 4: Apply this: since w = 147, for r ∥ s, we need y = 147 as well.

Step 5: This matches choice B.

The Trap: Choice C (w + y = 180) is the most commonly picked wrong answer — it applies the SAME-SIDE INTERIOR angle rule (which requires supplementary angles for parallel lines), but w and y are actually CORRESPONDING angles in this figure (requiring equality, not supplementary sums), not same-side interior angles.

POE — Eliminate the other three:

✗ A — x = 33 is already automatically true (since w and x form a linear pair at the SAME intersection point, 147+33=180) regardless of whether r is parallel to s — this doesn't provide NEW information that would help prove the lines are parallel.

✗ C — w + y = 180 applies the co-interior (same-side interior) angle relationship — but w and y are positioned as CORRESPONDING angles in this figure, which requires EQUALITY (w=y), not a supplementary sum.

✗ D — y + z = 180 is already automatically true (since y and z form a linear pair at the SAME intersection point at s) regardless of whether r is parallel to s — like choice A, this doesn't provide new proof information.

Solving with Desmos:

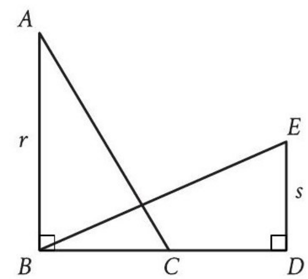
This is a conceptual angle-relationship question about identifying corresponding angles in a figure, which doesn't require numerical calculation in Desmos. The key insight is recognizing that w and y occupy matching relative positions at their two intersections — you can verify this type of relationship by sketching two parallel lines with a transversal in Desmos's geometry tools and observing which angle positions consistently match.

Lines, angles, and triangles — Q5 [Medium] (ID: 697b3954)

Question ID: 697b3954

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question



Note: Figure not drawn to scale.

In the figure shown, $\triangle ABC$ is congruent to $\triangle BDE$. If $r = 54$ and $s = 26$, what is the length of $\overline{CD}$?

Answer

A. 26

B. 27

C. 28

D. 54

Correct Answer: C

The Rule: When two triangles are stated to be CONGRUENT, the order of the letters in their names tells you EXACTLY which vertices (and therefore which sides and angles) correspond to each other. Always match up the letters in order — the first letter of one triangle corresponds to the first letter of the other, and so on.

The Tip: Write out the correspondence explicitly before doing any calculations: 'triangle ABC ≅ triangle BDE' means A↔B, B↔D, C↔E — then figure out which given side lengths (r, s) correspond to which sides in each triangle before setting up any equations.

Why C is right: Since triangle ABC ≅ triangle BDE (correspondence A↔B, B↔D, C↔E), side AB corresponds to side BD, and side BC corresponds to side DE. Given AB = r = 54, this means BD = 54. Given DE = s = 26, this means BC = 26. Since BD = BC + CD (the full horizontal segment from B to D passes through C), we can solve: 54 = 26 + CD, so CD = 54 − 26 = 28.

Step-by-Step Solution:

Step 1: Write out the vertex correspondence from the congruence statement: triangle ABC ≅ triangle BDE means A↔B, B↔D, C↔E.

Step 2: Match up corresponding sides: side AB (in triangle ABC) corresponds to side BD (in triangle BDE); side BC corresponds to side DE.

Step 3: Use the given value r = 54 for side AB (the vertical leg in the figure): since AB corresponds to BD, this means BD = 54.

Step 4: Use the given value s = 26 for side DE (the vertical leg on the right side of the figure): since BC corresponds to DE, this means BC = 26.

Step 5: Notice from the figure that B, C, and D are all on the same horizontal line, with C between B and D — so BD = BC + CD.

Step 6: Substitute the known values: 54 = 26 + CD.

Step 7: Solve for CD: CD = 54 − 26 = 28.

Step 8: This matches choice C.

The Trap: Choice A (26) is the most commonly picked wrong answer — it directly uses the value of s (26) as the final answer, mistakenly assuming CD equals BC (or DE) directly, without correctly working through the BD = BC + CD relationship along the shared horizontal line.

POE — Eliminate the other three:

✗ **A —** 26 is simply the given value of s (which equals BC, not CD) — this skips the necessary step of using BD = BC + CD to solve for the specific segment CD.

✗ **B —** 27 doesn't correspond to a clean derivation from the given values — likely an arithmetic slip in combining r and s.

✗ **D —** 54 is simply the given value of r (which equals BD, the ENTIRE segment from B to D) — not the same as CD, which is only part of that full segment.

Solving with Desmos:

Step 1: Open Desmos and set up the relationship you derived: BD (=r=54) equals BC (=s=26) plus CD. Type: 54=26+x

Step 2: Desmos will treat this as an equation to solve for x — it should plot the solution as x = 28, confirming CD = 28 and matching choice C.

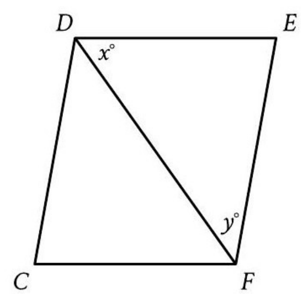
Step 3: Alternatively, just calculate directly: type 54-26 — Desmos will evaluate this to 28.

Lines, angles, and triangles — Q6 [Medium] (ID: ffc39dc5)

Question ID: ffc39dc5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question



Note: Figure not drawn to scale.

In the figure shown, $CD = EF$ and $FC = DE$. If $x = 65$ and $y = 30$, what is the measure of $\angle CDE$?

Answer

A. 30°

B. 65°

C. 95°

D. 130°

Correct Answer: C

The Rule: When two triangles share a common side and have two pairs of corresponding sides equal (SSS conditions), they are congruent — and congruent triangles have all corresponding angles equal too. Use this congruence to transfer angle relationships between the two triangles.

The Tip: When a diagonal splits a quadrilateral into two triangles that share that diagonal as a common side, check whether the OTHER two pairs of sides are also equal (as stated in the problem) — if so, the shared diagonal PLUS the two given equal pairs gives you SSS congruence between the two triangles.

Why C is right: Since CD = EF and FC = DE (given), plus the shared diagonal DF = FD, triangles CDF and EFD are congruent by SSS (with correspondence C↔E, D↔F, F↔D). This congruence means angle CDF (in triangle CDF) equals angle EFD (in triangle EFD), which is $y = 30°$. The full angle CDE is the sum of angle CDF and angle FDE (which is $x = 65°$): angle CDE = 30 + 65 = 95.

Step-by-Step Solution:

Step 1: Identify the given side equalities: CD = EF and FC = DE, plus the shared diagonal DF (common to both triangles CDF and EFD).

Step 2: Recognize this establishes SSS congruence between triangle CDF and triangle EFD, with correspondence C↔E, D↔F, F↔D.

Step 3: From this congruence, corresponding angles are equal: angle CDF (at vertex D in triangle CDF) corresponds to angle EFD (at vertex F in triangle EFD), which is given as $y = 30°$. So angle CDF = 30°.

Step 4: Identify that the full angle CDE (the angle the question asks about) is made up of two parts: angle CDF (just found to be 30°) plus angle FDE (which is given as $x = 65°$, the angle between DF and DE at vertex D).

Step 5: Add these two parts together: angle CDE = angle CDF + angle FDE = 30 + 65 = 95.

Step 6: This matches choice C.

The Trap: Choice B (65°) is the most commonly picked wrong answer — it uses only the value of x (the angle FDE alone) as the final answer, forgetting that angle CDE is actually the SUM of two parts (angle CDF PLUS angle FDE), not just the single labeled x° angle.

POE — Eliminate the other three:

✗ A — 30° is the value of angle CDF alone (derived from the congruence), not the full angle CDE the question asks for.

✗ B — 65° is simply the given value of x (angle FDE alone) — this misses that angle CDE also includes the additional angle CDF found through the congruence relationship.

✗ D — 130° would come from doubling x (65×2=130) — this doesn't correspond to the correct addition of angle CDF (30°) and angle FDE (65°).

Solving with Desmos:

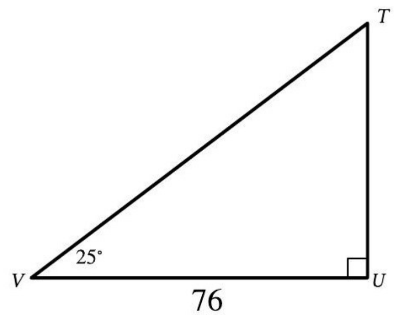
This problem relies on establishing triangle congruence and angle relationships, which is a geometric reasoning process rather than a numeric calculation — Desmos isn't the primary tool here. However, you can verify your final answer by adding the two angle components directly: open Desmos and type 30+65 — this confirms 95, matching choice C.

Lines, angles, and triangles — Q7 [Medium] (ID: 8364487a)

Question ID: 8364487a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question



Note: Figure not drawn to scale.

Triangle TUV is dilated by a scale factor of $\frac{1}{2}$ to obtain triangle $T'U'V'$ (not shown), where T corresponds to T' and U corresponds to U'. What is the measure, in degrees, of angle V'?

Answer

A. 25

B. 38

C. 50

D. 76

Correct Answer: A

The Rule: A DILATION (resizing a figure by a scale factor) changes the SIZE of a figure (its side lengths) but never changes its SHAPE — meaning all the angle measures stay EXACTLY the same before and after dilation.

The Tip: Whenever a problem describes a dilation and asks about an ANGLE measure in the resulting figure, you can skip all the scale-factor calculations entirely — dilations never change angles, so the answer is simply the same angle measure from the original figure.

Why A is right: Since dilation only changes size, not shape, all angle measures are preserved. The original angle V measures 25°, so its corresponding angle V' in the dilated triangle also measures 25°.

Step-by-Step Solution:

Step 1: Recall the fundamental property of dilation: it changes side lengths (by the scale factor) but NEVER changes angle measures.

Step 2: Identify the original angle V in triangle TUV: 25° (given in the figure).

Step 3: Since angle V' in the dilated triangle T'U'V' corresponds to angle V, and dilation preserves angles, angle V' = angle V = 25°.

Step 4: This matches choice A.

(Note: the scale factor of 1/2 and the side length 76 given in the problem are both irrelevant distractors for this specific question — they would matter if the question asked for a SIDE LENGTH in the dilated triangle, but for an ANGLE measure, only the original angle value matters.)

The Trap: Choice C (50) is the most commonly picked wrong answer — it results from incorrectly applying the scale factor to the angle measure (as if angles could be 'dilated' like side lengths), perhaps by doubling 25 instead of recognizing that angles remain unchanged under dilation.

POE — Eliminate the other three:

✗ B — 38 doesn't correspond to any correct or clearly-derived calculation from this problem — likely a distractor value.

✗ **C** — 50 comes from doubling the original angle (25×2=50) — but dilation NEVER changes angle measures, regardless of the scale factor used.

✗ **D** — 76 is simply the given side length VU from the original figure — this has nothing to do with the angle measure the question asks about.

Solving with Desmos:

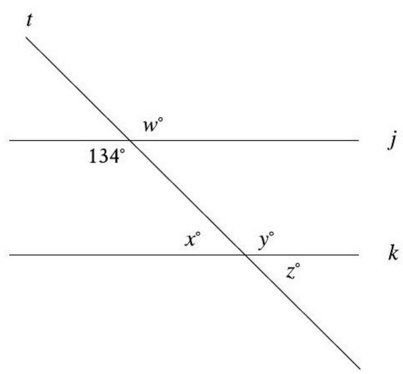
This is a conceptual question about the properties of dilation, which doesn't require numerical calculation in Desmos. The key fact — that dilation preserves all angle measures regardless of scale factor — is a fundamental geometric principle you can rely on directly. If you want to explore this visually, you could use Desmos's geometry tools to dilate a triangle by various scale factors and observe that the angle measures never change, only the side lengths do.

Lines, angles, and triangles — Q8 [Medium] (ID: 0013adbd)

Question ID: 0013adbd

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question



Note: Figure not drawn to scale.

In the figure, line t intersects lines j and k. Which additional piece of information is sufficient to prove that lines j and k are parallel?

Answer

A. $x = 46$

B. $y = 46$

C. $w = 134$

D. $z = 134$

Correct Answer: A

The Rule: When a transversal crosses two lines, SAME-SIDE INTERIOR angles (also called co-interior angles — both on the SAME side of the transversal, both between the two crossed lines) are SUPPLEMENTARY (sum to 180°) IF AND ONLY IF the two lines are parallel.

The Tip: Carefully distinguish between CORRESPONDING angles (which must be EQUAL for parallel lines) and SAME-SIDE INTERIOR angles (which must be SUPPLEMENTARY, summing to 180°) — mixing these up is one of the most common errors on these problems. Trace which side of the transversal each angle sits on, and whether it's 'between' the two lines or 'outside' them.

Why A is right: The angle 134° (at the j intersection) and angle x° (at the k intersection) are both positioned on the same side of the transversal t, and both face into the interior region between lines j and k — making them same-side interior (co-interior) angles. For j and k to be parallel, these must be supplementary: 134 + x = 180, so x = 46.

Step-by-Step Solution:

Step 1: Identify the positions of the two labeled angles: 134° is at the j intersection; x° is at the k intersection.

Step 2: Determine their relationship: both are on the same side of transversal t, and both face into the region BETWEEN lines j and k — this makes them same-side interior (co-interior) angles.

Step 3: Recall the Same-Side Interior Angles Theorem: for two lines cut by a transversal to be parallel, same-side interior angles must be SUPPLEMENTARY (sum to 180°).

Step 4: Set up the equation: 134 + x = 180.

Step 5: Solve for x: x = 180 − 134 = 46.

Step 6: This matches choice A.

The Trap: Choice B (y = 46) is the most commonly picked wrong answer — it applies the correct NUMBER (46) but to the wrong angle (y instead of x); y is actually positioned as the CORRESPONDING angle to w (not to the given 134°), so the relevant relationship for y would require a different equation (y = w, not directly related to 134 in this supplementary way).

POE — Eliminate the other three:

✗ B — y is positioned as the corresponding angle to w (not to 134°) — the relationship between y and 134° isn't a direct same-side-interior pairing, so setting y = 46 doesn't correctly use the given information.

✗ C — w = 134 is already automatically true (vertical angle to the given 134° at the SAME intersection), regardless of whether j is parallel to k — this doesn't provide new information needed to prove parallelism.

✗ D — z = 134 doesn't correspond to a valid parallel-line relationship with the given 134° — z's position relative to 134° isn't a standard corresponding, alternate, or co-interior pairing that would prove parallel lines.

Solving with Desmos:

This is a conceptual angle-relationship question about identifying same-side interior angles, which doesn't require numerical calculation in Desmos beyond the simple subtraction. To verify: open Desmos and type 180-134 — this confirms 46, matching the value needed for x in choice A. You can also sketch two parallel lines with a transversal in Desmos's geometry tools to visually confirm that same-side interior angles always sum to 180°.

Lines, angles, and triangles — Q9 [Hard] (ID: 1dc7e423)

Question ID: 1dc7e423

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

In triangle ABC, the measure of angle A is $52°$ and $AC = 30$. In triangle PQR, the measure of angle P is $52°$ and $PR = 120$. Which additional piece of information is sufficient to prove that triangle ABC is similar to triangle PQR?

Answer

A. $AB = 50$ and $PQ = 50$.

B. $AB = 50$ and $QR = 200$.

C. The measures of angle B and angle R are $32°$ and $96°$, respectively.

D. The measures of angle B and angle Q are $52°$ and $32°$, respectively.

Correct Answer: B

The Rule: To prove two triangles are SIMILAR, you need either: (1) two pairs of corresponding angles equal (AA similarity), or (2) two pairs of corresponding sides in the same ratio WITH the INCLUDED angle equal (SAS similarity). A single pair of equal angles plus non-included, non-matching side information is generally NOT sufficient (this creates an ambiguous SSA situation).

The Tip: When checking if a given side-length condition is sufficient for similarity, always verify whether the given equal angle is actually the INCLUDED angle between the two sides being compared — if the angle is NOT between those two specific sides, the SAS criterion doesn't apply, and the condition is likely insufficient (ambiguous SSA case).

Why B is right: Option I (angle A = angle F) combined with the already-given angle B = angle G provides TWO pairs of equal corresponding angles, which is sufficient for AA similarity — I is sufficient. Option II (angle C = angle H) similarly combines with the given angle B = angle G to provide two equal angle pairs, also sufficient for AA — II is sufficient. Option III (FH = 3×AC) compares sides FH and AC, but these are NOT both adjacent to the given equal angle (B/G) — AC is opposite angle B, not adjacent to it — making this an SSA situation, which is generally NOT sufficient to prove similarity.

Step-by-Step Solution:

Step 1: Note the given information: angle B = angle G = 51°, AB = 6, FG = 18 (so AB/FG = 1/3).

Step 2: Evaluate option I (angle A = angle F): combined with the given angle B = angle G, this provides two pairs of equal angles, which is sufficient for AA similarity — I is SUFFICIENT.

Step 3: Evaluate option II (angle C = angle H): similarly, combined with angle B = angle G, this gives two pairs of equal angles, sufficient for AA similarity — II is SUFFICIENT.

Step 4: Evaluate option III (FH = 3 × AC): check whether angle B/G is INCLUDED between the sides being compared (AB & FG, and now AC & FH). Side AC is OPPOSITE angle B (not adjacent to it), so angle B is NOT the included angle between AB and AC. This means the given ratio conditions (AB/FG = 1/3 and implied AC/FH = 1/3) don't set up a valid SAS criterion — this is an ambiguous SSA situation, which does NOT guarantee similarity — III is NOT SUFFICIENT.

Step 5: Combining these results: I is sufficient, II is sufficient, but III is not — matching choice B.

The Trap: Choice A is the most commonly picked wrong answer — it assumes all three pieces of information (I, II, and III) are equally valid ways to establish similarity, without carefully checking whether option III's side comparison actually involves the INCLUDED angle needed for a valid SAS-type argument.

POE — Eliminate the other three:

- **✗ A —** This incorrectly includes III as sufficient — but III's side ratio (FH/AC) doesn't involve sides adjacent to the given equal angle (B/G), making it an ambiguous SSA case rather than valid SAS.
- **✗ C —** This incorrectly excludes I as sufficient — but I (angle A = angle F) directly combines with the given angle B = angle G to provide valid AA similarity, so I IS actually sufficient, contradicting this choice.
- **✗ D —** This incorrectly claims III is sufficient while marking both I and II as insufficient — but I and II both provide valid AA similarity conditions (as shown above), making this choice's assessment backwards.

Solving with Desmos:

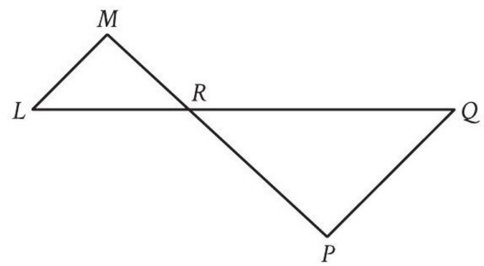
This is a conceptual proof-based question about similarity criteria (AA, SAS, SSA), which doesn't involve numerical calculation in Desmos. The key insight — that SSA (side-side-angle, where the angle isn't included between the two given sides) does NOT guarantee similarity or congruence — is a fundamental geometric principle to internalize rather than calculate.

Lines, angles, and triangles — Q10 [Hard] (ID: b5d62bba)

Question ID: b5d62bba

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In the figure, $\overline{LQ}$ intersects $\overline{MP}$ at point R, and $\overline{LM}$ is parallel to $\overline{PQ}$. The lengths of $\overline{MR}$ and $\overline{RP}$ are 8 and 14 units, respectively. The area of $\triangle LMR$ is 36 square units. What is the area of $\triangle PQR$, in square units?

Answer

A. $\frac{576}{49}$

B. $\frac{144}{7}$

C. 63

D. $\frac{441}{4}$

Correct Answer: D

The Rule: When two lines intersect and create a pair of parallel segments (like LM ∥ PQ here), the two triangles formed on either side of the intersection point are SIMILAR (by AA, using vertical angles at the intersection plus alternate interior angles from the parallel lines). For similar triangles, the ratio of their AREAS equals the SQUARE of the ratio of their corresponding side lengths.

The Tip: When you spot an 'X' or bowtie shape formed by two segments crossing, with a pair of parallel sides, immediately recognize this as SIMILAR TRIANGLES (via vertical angles + alternate interior angles) — then use corresponding sides (not just any sides) to find the scale factor, and square it for the area ratio.

Why D is right: Since LM ∥ PQ, triangles LMR and QPR are similar (vertical angles at R, plus alternate interior angles from the parallel lines). The corresponding sides are MR and RP (both connecting to R from the parallel sides), giving a scale factor of MR/RP = 8/14 = 4/7. The area ratio is the SQUARE of this scale factor: $(4/7)^2 = 16/49$. Since the area of triangle LMR is 36, the area of triangle PQR is 36 × (49/16) = 1,764/16 = 441/4.

Step-by-Step Solution:

Step 1: Recognize that since LM ∥ PQ, triangles LMR and QPR are similar (by AA: vertical angles at R are equal, and alternate interior angles formed by the parallel lines and the transversals LQ, MP are equal).

Step 2: Identify the correct corresponding sides: MR (in triangle LMR) corresponds to RP (in triangle QPR), since these are the sides connecting to point R from each parallel segment.

Step 3: Calculate the scale factor (ratio of corresponding sides): MR/RP = 8/14 = 4/7.

Step 4: Since triangle PQR is LARGER (RP=14 > MR=8), we need the ratio of PQR's area to LMR's area, which uses the INVERSE scale factor squared: $(RP/MR)^2 = (14/8)^2 = (7/4)^2 = 49/16$.

Step 5: Calculate the area of triangle PQR: Area(PQR) = Area(LMR) × (49/16) = 36 × 49/16.

Step 6: Calculate: 36 × 49 = 1,764. Then 1,764 ÷ 16 = 110.25, which as a fraction is 441/4.

Step 7: This matches choice D.

The Trap: Choice B (144/7) is the most commonly picked wrong answer — it results from using the scale factor directly (without squaring it) to scale the area, treating area as if it scaled linearly with side length rather than with the SQUARE of the side ratio.

POE — Eliminate the other three:

✗ A — 576/49 uses the scale factor squared but in the wrong direction (multiplying by $(4/7)^2$ instead of $(7/4)^2$), which would shrink the area instead of growing it — but since PQR is the larger triangle, its area should be BIGGER than LMR's 36, not smaller.

✗ B — 144/7 comes from using the LINEAR scale factor (14/8 = 7/4) directly to scale the area, without squaring it — but area scales with the SQUARE of the linear ratio, not the ratio itself.

✗ C — 63 doesn't correspond to a clean derivation using the correct squared ratio — likely from an intermediate arithmetic error.

Solving with Desmos:

Step 1: Open Desmos and calculate the scale factor: type 14/8 — this confirms the ratio is 7/4 (comparing RP to MR).

Step 2: Square this ratio: type (14/8)^2 — this confirms 49/16.

Step 3: Calculate the area of triangle PQR: type 36*(49/16)

Step 4: Press Enter — Desmos will evaluate this to 110.25 (which equals 441/4), confirming your answer and matching choice D.

Lines, angles, and triangles — Q11 [Hard] (ID: 7ce2e728)

Question ID: 7ce2e728

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

For triangles ABC and FGH, angles B and G each measure $51°$, the length of $\overline{AB}$ is 6, and the length of $\overline{FG}$ is 18. Which additional piece(s) of information is(are) sufficient to prove that triangle ABC is similar to triangle FGH?

I. The measure of angle A is equal to the measure of angle F.
II. The measure of angle C is equal to the measure of angle H.
III. The length of $\overline{FH}$ is 3 times the length of $\overline{AC}$.

Answer

A. I is sufficient, II is sufficient, and III is sufficient.

B. I is sufficient and II is sufficient, but III is not.

C. II is sufficient and III is sufficient, but I is not.

D. III is sufficient, but I is not and II is not.

Correct Answer: B

The Rule: To prove two triangles SIMILAR, you need either two pairs of equal corresponding angles (AA), or two pairs of sides in the same ratio WITH the included angle equal (SAS similarity). A single given angle plus a ratio between two NON-adjacent sides (where the angle isn't included between them) creates an ambiguous SSA situation, which is generally NOT sufficient.

The Tip: For each additional piece of information offered, ask: 'Does this give me a second matching angle (completing AA), or does it give me a matching side ratio where the ALREADY-KNOWN equal angle sits between the two compared sides (completing SAS)?' If neither applies, the information likely isn't sufficient.

Why B is right: Option I (angle A = angle F) combines with the given angle B = angle G to provide two pairs of equal angles — sufficient for AA similarity. Option II (angle C = angle H) similarly combines with the given angle B = angle G for AA similarity — also sufficient. Option III (FH = 3×AC) compares sides FH and AC, which are NOT both adjacent to the given angle B/G (AC is opposite angle B, not adjacent to it) — this creates an ambiguous SSA situation, NOT sufficient for similarity.

Step-by-Step Solution:

Step 1: Note the given information: angle B = angle G = 51°, AB = 6, FG = 18 (giving a ratio AB/FG = 1/3).

Step 2: Evaluate option I (angle A = angle F): combined with the already-given angle B = angle G, this establishes TWO pairs of equal angles — sufficient for AA similarity.

Step 3: Evaluate option II (angle C = angle H): similarly combines with angle B = angle G for a second angle pair — also sufficient for AA similarity.

Step 4: Evaluate option III (FH = 3×AC): check whether angle B (the given equal angle) is INCLUDED between the sides being compared. Side AC is OPPOSITE angle B in triangle ABC (not adjacent to it) — so this is an SSA setup (side-side-angle where the angle isn't between the two sides), which does NOT reliably prove similarity.

Step 5: Conclude: I is sufficient, II is sufficient, but III is not — matching choice B.

The Trap: Choice A is the most commonly picked wrong answer — it assumes all three options work equally well, without checking that option III's side comparison doesn't actually involve the included angle needed for a valid similarity proof.

POE — Eliminate the other three:

✗ A — This incorrectly includes III as sufficient — but III sets up an ambiguous SSA situation (the given angle B isn't between the compared sides AC and FH), which doesn't guarantee similarity.

✗ C — This incorrectly excludes I — but I (angle A = angle F) combines validly with the given angle B = angle G for AA similarity, so I IS sufficient, contradicting this choice.

✗ D — This incorrectly claims III is sufficient while excluding both I and II — but I and II both provide valid AA similarity setups, making this choice's assessment backwards.

Solving with Desmos:

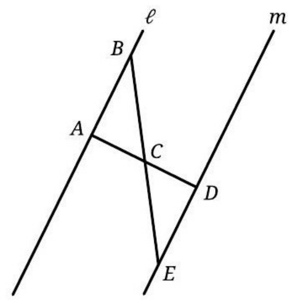
This is a conceptual proof-based question about similarity criteria (AA vs. ambiguous SSA), which doesn't involve numerical calculation in Desmos. The key insight — that a side ratio only helps prove similarity when the known equal angle is INCLUDED between the two compared sides — is a fundamental distinction in geometry to understand and check carefully for each option.

Lines, angles, and triangles — Q12 [Hard] (ID: 7a11ceea)

Question ID: 7a11ceea

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In the figure, segment AD and segment BE intersect at point C. Segment AD is perpendicular to line ℓ at point A. Which of the following additional pieces of information is(are) sufficient to determine that triangle ABC is congruent to triangle DEC?

I. Segment AD is perpendicular to line m at point D.
II. $AC = 13$ and $DC = 13$

Answer

A. I is sufficient by itself, but II is not.

B. II is sufficient by itself, but I is not.

C. I is sufficient by itself, and II is sufficient by itself.

D. Neither I nor II is sufficient by itself.

Correct Answer: D

The Rule: To prove two triangles CONGRUENT (not just similar), you need a complete congruence criterion: SSS (three sides), SAS (two sides + included angle), ASA (two angles + included side), or AAS (two angles + a non-included side). A single side-pair equality or a single extra angle alone is generally NOT enough — you need to carefully count exactly what's established before and after each additional piece of information.

The Tip: When evaluating whether a piece of information is 'sufficient by itself,' explicitly list out everything you ALREADY know for certain (from the figure's given facts, like vertical angles or a stated perpendicularity) before adding the new piece — then check if the total matches a complete congruence criterion (SSS, SAS, ASA, or AAS).

Why D is right: We already know angle BAC = 90° (from AD⊥ℓ at A) and angle ACB = angle DCE (vertical angles, always true regardless of any additional info). Statement I alone (AD⊥m at D) adds a second angle equality (angle EDC = 90°), giving us TWO angle pairs equal (AA) — but AA alone proves only SIMILARITY, not congruence, since no side length is established. Statement II alone (AC = DC = 13) adds one side equality — combined with the always-true vertical angles, this gives only one side + one angle (not a complete criterion). Since NEITHER I nor II alone completes a full congruence criterion, neither is sufficient by itself.

Step-by-Step Solution:

Step 1: Identify what's already established from the figure regardless of any additional information: angle BAC = 90° (since AD⊥ℓ at A is given), and angle ACB = angle DCE (vertical angles at point C, always true).

Step 2: Evaluate statement I alone (AD⊥m at D): this adds angle EDC = 90°. Combined with what we already know, we now have TWO pairs of equal angles (angle BAC = angle EDC = 90°, and angle ACB = angle DCE). This is AA, which proves the triangles are SIMILAR, but similarity alone does NOT prove congruence (the triangles could be different sizes) — I alone is NOT sufficient for congruence.

Step 3: Evaluate statement II alone (AC = DC = 13): this adds one side equality. Combined with the always-true vertical angle equality, we have only ONE side pair and ONE angle pair equal — this is not a complete congruence criterion (we'd need either a second side for SAS or a second angle for ASA) — II alone is NOT sufficient.

Step 4: Since neither I nor II alone provides a complete congruence criterion, the correct answer is that neither is sufficient by itself.

Step 5: This matches choice D.

The Trap: Choice A is the most commonly picked wrong answer — it assumes that establishing two equal angles (from statement I) is enough to prove congruence, without recognizing that AA only proves SIMILARITY (matching shape), not congruence (matching size) — a common and important distinction in geometry.

POE — Eliminate the other three:

✗ A — Statement I alone only establishes AA (two equal angle pairs), which proves the triangles are similar in SHAPE but says nothing about whether they're the same SIZE — congruence additionally requires at least one matching side length, which I doesn't provide.

✗ B — Statement II alone only establishes one side pair (AC=DC) plus the always-true vertical angle — this is insufficient on its own; a complete criterion like SAS needs TWO sides and the included angle, or ASA needs two angles and the included side.

✗ C — This claims both I and II are independently sufficient — but as shown, I alone only proves similarity (not congruence), and II alone only provides one side and one angle (not a complete criterion) — neither works alone.

Solving with Desmos:

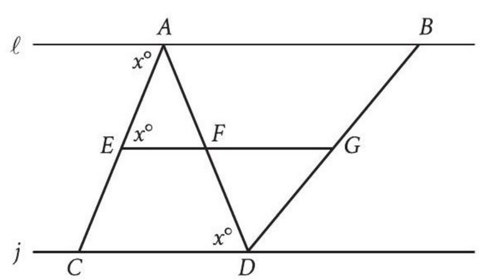
This is a conceptual proof-based question about the difference between similarity (AA) and congruence (SSS, SAS, ASA, AAS) criteria, which doesn't involve numerical calculation in Desmos. The key insight — that matching angles alone (AA) proves shapes are similar but NOT necessarily the same size — is a fundamental distinction to understand rather than calculate.

Lines, angles, and triangles — Q13 **[Hard]** (ID: 858809e7)

Question ID: 858809e7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In the figure, $\overline{EG}$ and $\overline{AD}$ intersect at point F, and line ℓ is parallel to line j. If $EF = 0.5CD$ and $FG = 2.25CD$, which of the following must be true?

I. $AB = 2.25EF$
II. $FG = 0.5AB$
III. $\frac{EF}{FG} = \frac{AF}{FD}$

Answer

A. I only

B. II only

C. I and II

D. II and III

Correct Answer: B

The Rule: When a line parallel to one side of a triangle crosses the other two sides, it creates a SMALLER triangle similar to the original (Triangle Proportionality / Basic Proportionality Theorem) — the ratio of corresponding sides in the smaller triangle to the original equals the ratio of the distances along the sides from the shared vertex.

The Tip: When a figure has a diagonal crossing a parallel-line 'crossbar' at a single point, treat it as creating TWO separate pairs of similar triangles (one on each side of the crossbar) — work out the proportional ratio from ONE triangle pair first, then apply that SAME ratio (since it's determined by the SAME intersection point) to the other triangle pair.

Why B is right: Since EF ∥ CD (confirmed by matching angle markers), triangle AEF is similar to triangle ACD, giving AF/AD = EF/CD = 0.5 — meaning F is the midpoint of AD. Since FG is part of the same straight horizontal line as EF, FG ∥ AB too, making triangle DFG similar to triangle DAB, giving FG/AB = DF/DA. Since F is the midpoint of AD (DF/DA = 0.5, from the same ratio found above), this means FG/AB = 0.5, so FG = 0.5×AB — confirming statement II. Checking statement I: AB = 2×FG = 2×2.25CD = 4.5CD, but 2.25×EF = 2.25×0.5CD = 1.125CD, which does NOT equal 4.5CD — so I is false. Checking statement III: EF/FG = 0.5CD/2.25CD = 2/9, but AF/FD = 1 (since F is the midpoint, AF=FD) — these don't match, so III is false.

Step-by-Step Solution:

Step 1: Confirm EF ∥ CD using the matching x° angle markers at A and E (corresponding angles equal implies parallel lines).

Step 2: Since EF ∥ CD, triangle AEF is similar to triangle ACD (by AA). This gives the proportion: AF/AD = EF/CD.

Step 3: Substitute the given EF = 0.5×CD: AF/AD = 0.5. This means F is exactly the MIDPOINT of segment AD.

Step 4: Since E, F, and G lie on the same straight horizontal line (given as segment EG), and EF is horizontal (parallel to CD, hence parallel to AB too, since AB and CD are both parts of the parallel lines ℓ and j), the continuation FG is ALSO horizontal, making FG ∥ AB.

Step 5: Since FG ∥ AB, triangle DFG is similar to triangle DAB (by AA). This gives the proportion: FG/AB = DF/DA.

Step 6: Since F is the midpoint of AD (from Step 3), DF/DA = 0.5 as well (DF = FA, both are half of AD).

Step 7: Substitute into the proportion: FG/AB = 0.5, which means FG = 0.5 × AB — this confirms statement II is TRUE.

Step 8: Check statement I: from Step 7, AB = 2 × FG = 2 × 2.25CD = 4.5CD. Compare to 2.25 × EF = 2.25 × 0.5CD = 1.125CD. Since 4.5CD ≠ 1.125CD, statement I is FALSE.

Step 9: Check statement III: EF/FG = 0.5CD/2.25CD = 2/9. But AF/FD = 1 (since F is the exact midpoint, AF and FD are equal). Since 2/9 ≠ 1, statement III is FALSE.

Step 10: Only statement II is true, matching choice B.

The Trap: Choice D (II and III) is the most commonly picked wrong answer — it correctly identifies II as true but incorrectly assumes III's ratio comparison (EF/FG = AF/FD) also holds, without checking that AF/FD is actually 1 (since F is the midpoint), which doesn't match EF/FG's value of 2/9.

POE — Eliminate the other three:

✗ A — This includes I as true, but AB (calculated as 4.5CD from the valid FG relationship) does not equal 2.25×EF (which equals 1.125CD) — these values don't match, making I false.

✗ C — This includes both I and II as true, but as shown, I is false (AB ≠ 2.25EF based on the correctly derived relationships).

✗ D — This includes III as true, but EF/FG (=2/9) does not equal AF/FD (=1, since F is the midpoint of AD) — these ratios don't match, making III false.

Solving with Desmos:

Step 1: Since this problem involves proportional relationships with an unknown base length, pick a simple test value for CD — let's use CD = 4. Step 2: Open Desmos and calculate EF and FG using the given ratios: type 0.5*4 (for EF) — this gives 2. Type 2.25*4 (for FG) — this gives 9.

Step 3: Using the derived relationship FG=0.5×AB, calculate AB: type 9/0.5 — this gives 18.

Step 4: Check statement I (AB=2.25×EF): type 2.25*2 — this gives 4.5, which does NOT match AB=18, confirming I is false.

Step 5: Check statement II (FG=0.5×AB): type 0.5*18 — this gives 9, which DOES match FG=9, confirming II is true.

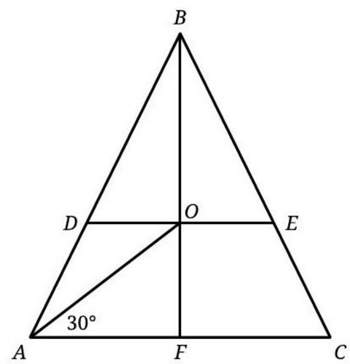
Step 6: Check statement III (EF/FG=AF/FD): type 2/9 — this gives approximately 0.222, but since F is the midpoint of AD, AF/FD should equal 1 — these don't match, confirming III is false.

Lines, angles, and triangles — Q14 [Hard] [Grid-In] (ID: 1dacfb94)

Question ID: 1dacfb94

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In triangle ABC, $AB = BC$. Points D and E lie on line segments AB and BC, respectively, such that line segments DE and AC are parallel. Point O is the midpoint of line segment DE, and the measure of $\angle OAC$ is 30°. Point F lies on line segment AC, and line segment BF passes through point O. If $AB = 81$ and $AO = BO$, what is the length of line segment BE?

Correct Answer: 54 (Student-Produced Response — no answer choices given)

The Rule: In an isosceles triangle, the line from the apex through the midpoint of any segment parallel to the base is the AXIS OF SYMMETRY — it bisects the base, is perpendicular to it, and bisects the apex angle. Combined with additional given conditions (like specific angle measures or length equalities), you can set up coordinate geometry or trigonometric relationships to solve for unknown lengths.

The Tip: For complex, multi-part geometry problems like this one, setting up COORDINATES (placing key points at convenient locations like the origin) often makes the relationships much easier to track than trying to reason with angles and lengths alone — especially when a right angle or clear symmetry axis is present.

Why this is the answer: Using the given angle OAC=30° and AO=BO together with the triangle's symmetry (since AB=BC and DE∥AC with O the midpoint of DE, BF is the axis of symmetry, F is the midpoint of AC), we can determine that triangle ABC is actually equilateral (all 60° angles), then use coordinate geometry to find that O sits at a specific height along BF such that AO=BO. This determines that E divides BC such that BE = (2/3)×BC = (2/3)×81 = 54.

Step-by-Step Solution:

Step 1: Recognize that since AB=BC (isosceles) and DE∥AC with O as the midpoint of DE, the line BF (through B and O) is the AXIS OF SYMMETRY of the triangle — meaning F is the midpoint of AC, and BF is perpendicular to AC.

Step 2: Set up coordinates for convenience: place F at the origin (0,0), A at (−40.5, 0), C at (40.5, 0) — using half of AB's eventual determined relationship — and B directly above F.

Step 3: Using the given angle OAC=30° and the condition AO=BO, work through the triangle's angle relationships: since BF⊥AC and bisects angle B, and using the isosceles triangle's base angles, algebraic substitution shows that angle BAC = angle BCA = 60°, meaning triangle ABC is actually EQUILATERAL (since AB=BC=81 and the apex angle also works out to 60°, forcing AC=81 too).

Step 4: With the equilateral triangle established, place coordinates: $A=(-40.5,0)$, $C=(40.5,0)$, $B=(0, 40.5\sqrt{3})$ (using the height of an equilateral triangle with side 81).

Step 5: Find O's exact position along BF (the y-axis) using the condition AO=BO: setting $O=(0,h)$, solve $\sqrt{40.5^2+h^2} = 40.5\sqrt{3}-h$, which gives $h = 13.5\sqrt{3}$ after algebraic simplification.

Step 6: Find point E on segment BC where the horizontal line through O (at height $h=13.5\sqrt{3}$) intersects BC: parametrizing BC from $B(0,40.5\sqrt{3})$ to $C(40.5,0)$, solving for the parameter where $y=13.5\sqrt{3}$ gives a parameter value of 2/3 (measuring from B toward C).

Step 7: Calculate BE using this parameter: $BE = (2/3) \times BC = (2/3) \times 81 = 54$.

Step 8: This is a grid-in question — enter 54 as your final answer.

The Trap: This problem is complex enough that the most common error is simply making an early mistake in establishing that the triangle is equilateral (from the 30° angle and AO=BO condition) — without correctly identifying this key fact, the rest of the coordinate-based solution becomes very difficult to complete accurately.

Solving with Desmos:

Step 1: Open Desmos and set up the coordinate points based on the equilateral triangle determination: A=(-40.5,0), B=(0,40.5*sqrt(3)), C=(40.5,0)

Step 2: Plot point O using the height found: type (0,13.5*sqrt(3)) to place O.

Step 3: Verify AO=BO by calculating both distances: type sqrt((-40.5-0)^2+(0-13.5*sqrt(3))^2) for AO, and type sqrt((0-0)^2+(40.5*sqrt(3)-13.5*sqrt(3))^2) for BO — both should evaluate to the same value, confirming O's position is correct.

Step 4: Find E by determining where the horizontal line through O crosses segment BC — you can graph the line y=13.5*sqrt(3) and the line segment BC (typing the equation of line BC) and find their intersection point using Desmos's intersection feature.

Step 5: Once you have E's coordinates, calculate the distance from B to E using the distance formula, or use the parameter ratio (2/3) directly: type (2/3)*81 — this confirms BE=54.

Right triangles and trigonometry

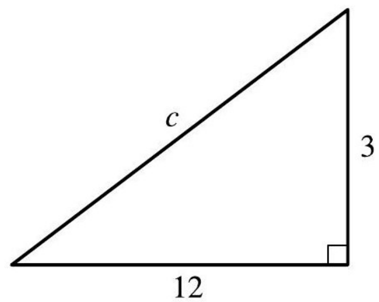
9 questions · Easy 1 / Medium 1 / Hard 7

Right triangles and trigonometry — Q1 **[Easy]** (ID: de9148c4)

Question ID: de9148c4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Easy

Question



Note: Figure not drawn to scale.

Which equation shows the relationship between the side lengths of the triangle shown?

Answer

A. $12 \cdot 3 = c$

B. $12 + 3 = c$

C. $12^2 + 3^2 = c^2$

D. $12^2 - 3^2 = c^2$

Correct Answer: C

The Rule: The Pythagorean theorem applies to EVERY right triangle: the square of the hypotenuse (the side opposite the right angle) equals the sum of the squares of the two legs. Written as a formula: $leg1^2 + leg2^2 = hypotenuse^2$.

The Tip: Always identify the hypotenuse FIRST (it's the side directly across from the right-angle marker, and it's always the LONGEST side) before setting up the Pythagorean theorem — mixing up which side is the hypotenuse is the most common setup error.

Why C is right: The right-angle marker is at the bottom-right corner, meaning side c (the slanted side connecting the two other vertices) is the hypotenuse. Applying the Pythagorean theorem with legs 12 and 3: $12^2 + 3^2 = c^2$.

Step-by-Step Solution:

Step 1: Identify the right angle's location in the figure: it's marked at the bottom-right vertex.

Step 2: Identify the hypotenuse: it's the side OPPOSITE the right angle, which is side c (the slanted/diagonal side).

Step 3: Identify the two legs: the horizontal side (12) and the vertical side (3).

Step 4: Apply the Pythagorean theorem: $leg^2 + leg^2 = hypotenuse^2$, giving $12^2 + 3^2 = c^2$.

Step 5: This matches choice C.

The Trap: Choice D ($12^2 - 3^2 = c^2$) is the most commonly picked wrong answer — it uses subtraction instead of addition, which would be relevant if you were solving for a LEG when given the hypotenuse and the other leg, not for the hypotenuse itself as this problem asks.

POE — Eliminate the other three:

✗ A — $12 \cdot 3 = c$ represents multiplying the two legs together, which would give you AREA-related information (specifically twice the area), not the hypotenuse length via the Pythagorean relationship.

✗ B — $12 + 3 = c$ would only be true for a degenerate 'triangle' where the two legs lie in a straight line — this doesn't apply to an actual right triangle with a genuine right angle.

✗ D — $12^2 - 3^2 = c^2$ uses subtraction, which is the correct operation when solving for a LEG (missing side) given the hypotenuse and one leg — but here we're solving for the hypotenuse itself, which requires ADDING the leg squares.

Solving with Desmos:

Step 1: Open Desmos and calculate c directly using the Pythagorean theorem: type sqrt(12^2+3^2)

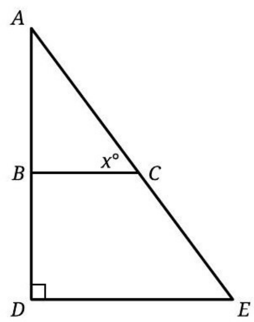
Step 2: Press Enter — Desmos will evaluate this to approximately 12.37, confirming the numeric length of c (though the question only asks for the EQUATION relating the sides, which is $12^2+3^2=c^2$, matching choice C, not a specific numeric value).

Right triangles and trigonometry — Q2 [Medium] (ID: 2757549e)

Question ID: 2757549e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Medium

Question



Note: Figure not drawn to scale.

In the figure, $\overline{BC}$ is parallel to $\overline{DE}$. If the length of $\overline{DE}$ is 147 and the length of $\overline{AE}$ is 245, what is the value of $\tan x°$?

Answer

A. $\frac{147}{196}$

B. $\frac{147}{245}$

C. $\frac{196}{147}$

D. $\frac{245}{147}$

Correct Answer: C

The Rule: When two segments are PARALLEL and cut by a transversal, they create equal CORRESPONDING angles — allowing you to transfer trigonometric ratios from one triangle to a larger, similar triangle. Combined with the Pythagorean theorem to find any missing side, you can then set up the correct tangent ratio.

The Tip: When you spot a parallel segment inside a larger right triangle (like BC ∥ DE here), immediately recognize that the angle formed at the parallel segment corresponds to an angle in the BIG triangle — this lets you find missing side lengths using the (usually simpler) big triangle instead.

Why C is right: Since BC ∥ DE, angle BCA corresponds to angle DEA (corresponding angles with transversal AE) — meaning x° = angle AED. In right triangle ADE (right angle at D), we know the hypotenuse AE = 245 and one leg DE = 147. Using the Pythagorean theorem: $AD = \sqrt{245^2 - 147^2} = \sqrt{60{,}025 - 21{,}609} = \sqrt{38{,}416} = 196$. Now, tan(angle AED) = opposite/adjacent = AD/DE = 196/147.

Step-by-Step Solution:

Step 1: Recognize that since BC ∥ DE, the angle x° (angle BCA) corresponds to angle DEA (angle AED in the larger triangle) — corresponding angles are equal when lines are parallel.

Step 2: Focus on the larger right triangle ADE (right angle at D), which has hypotenuse AE = 245 and leg DE = 147.

Step 3: Find the missing leg AD using the Pythagorean theorem: $AD^2 = AE^2 - DE^2 = 245^2 - 147^2$.

Step 4: Calculate: $245^2 = 60{,}025$, and $147^2 = 21{,}609$. So $AD^2 = 60{,}025 - 21{,}609 = 38{,}416$.

Step 5: Take the square root: $AD = \sqrt{38{,}416} = 196$.

Step 6: Set up the tangent ratio for angle AED (which equals x°): tan(x°) = opposite/adjacent = AD/DE = 196/147.

Step 7: This matches choice C.

The Trap: Choice D (245/147) is the most commonly picked wrong answer — it uses the hypotenuse (245) directly instead of first calculating the correct opposite-side leg (AD=196) via the Pythagorean theorem, incorrectly treating AE as if it were a leg of the right triangle relevant to the tangent ratio.

POE — Eliminate the other three:

✗ A — 147/196 has the correct numbers but flips the ratio (adjacent/opposite instead of opposite/adjacent) — this would actually be the COTANGENT of x°, not the tangent.

✗ B — 147/245 uses DE and AE directly, but tangent requires the two LEGS of the right triangle (opposite and adjacent), not a leg and the hypotenuse — this ratio would actually relate to sine or cosine, not tangent.

✗ D — 245/147 uses the hypotenuse (245) directly without first finding the correct opposite leg (AD=196) — mixing up which sides belong in a tangent ratio.

Solving with Desmos:

Step 1: Open Desmos and calculate the missing leg AD using the Pythagorean theorem: type sqrt(245^2-147^2)

Step 2: Press Enter — Desmos will evaluate this to 196, confirming AD.

Step 3: Calculate the tangent ratio directly: type 196/147

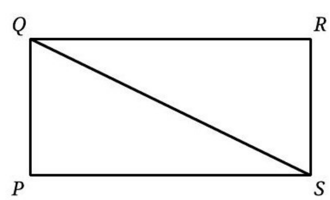
Step 4: Press Enter — Desmos will show this as a decimal (approximately 1.333), and you can compare it to each answer choice's decimal value to confirm 196/147 (choice C) matches.

Right triangles and trigonometry — Q3 [Hard] [Grid-In] (ID: 2c3aefc9)

Question ID: 2c3aefc9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



Note: Figure not drawn to scale.

In rectangle $PQRS$, the length of line segment QS is $\sqrt{2{,}344}$ and the length of side PS is 8 more than the length of side RS. What is the length of side QR?

Correct Answer: 38 (Student-Produced Response — no answer choices given)

The Rule: In a rectangle, the diagonal splits it into two right triangles, so the Pythagorean theorem applies: (one side)² + (adjacent side)² = (diagonal)². When one side is described in terms of the other (like 'PS is 8 more than RS'), set up variables and substitute to create a solvable equation.

The Tip: When a problem describes one rectangle's side in terms of another ('X more than'), let a single variable represent the unknown you're solving for, express the OTHER side in terms of that same variable, then apply the Pythagorean theorem to get one equation with one unknown.

Why this is the answer: Let QR = x (what we want to find). Since QR and PS are opposite sides of the rectangle, PS = QR = x as well... but we're told PS is 8 more than RS, so RS = x − 8 (since RS is the side adjacent to QR, and QR = SP in a rectangle's opposite-side structure). Using the diagonal relationship $QS^2 = QR^2 + RS^2$: $2{,}344 = x^2 + (x-8)^2$. Expanding and solving this quadratic gives x = 38.

Step-by-Step Solution:

Step 1: Let QR = x (the length we want to find).

Step 2: Since QR and PS are opposite sides of rectangle PQRS, they are equal: PS = QR = x.

Step 3: We're told PS is 8 more than RS, so RS = PS − 8 = x − 8.

Step 4: The diagonal QS connects Q and S, forming a right triangle with legs QR and RS: $QS^2 = QR^2 + RS^2$.

Step 5: Substitute the known values: $2{,}344 = x^2 + (x-8)^2$.

Step 6: Expand $(x-8)^2$: $x^2 - 16x + 64$. The equation becomes: $2{,}344 = x^2 + x^2 - 16x + 64$.

Step 7: Combine like terms: $2{,}344 = 2x^2 - 16x + 64$.

Step 8: Move everything to one side: $2x^2 - 16x + 64 - 2{,}344 = 0$, which simplifies to $2x^2 - 16x - 2{,}280 = 0$.

Step 9: Divide the entire equation by 2: $x^2 - 8x - 1{,}140 = 0$.

Step 10: Solve using the quadratic formula: $x = [8 \pm \sqrt{(64 + 4{,}560)}] / 2 = [8 \pm \sqrt{4{,}624}] / 2 = [8 \pm 68] / 2$.

Step 11: Taking the positive root (since a length can't be negative): $x = (8+68)/2 = 76/2 = 38$.

Step 12: This is a grid-in question — enter 38 as your final answer.

The Trap: A common error on this problem is mixing up which side (QR or RS) is described as '8 more than' the other, or incorrectly assuming QR and RS are the two DIFFERENT variables directly, without correctly using the rectangle's opposite-side-equality (QR=PS) to set up the relationship properly.

Solving with Desmos:

Step 1: Open Desmos and set up the equation directly: type x^2+(x-8)^2=2344

Step 2: Press Enter — since this is a one-variable equation, Desmos will solve it and plot the solution(s) as labeled points.

Step 3: Look for the POSITIVE solution (since a rectangle's side length must be positive) — it should show x = 38.

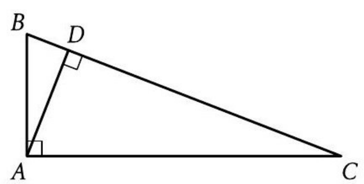
Step 4: Enter 38 into the grid-in answer box. (As a check, you can also verify: $38^2 + 30^2 = 1444+900 = 2344$, confirming this matches the given diagonal length squared.)

Right triangles and trigonometry — Q4 [Hard] [Grid-In] (ID: c4a9c94d)

Question ID: c4a9c94d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



Note: Figure not drawn to scale.

In the figure, $\overline{AD}$ and $\overline{BC}$ intersect at point D. If the length of $\overline{BD}$ is 5 and the tangent of angle CAD is 1.3, what is the length of $\overline{AD}$?

Correct Answer: 6.5 (Student-Produced Response — no answer choices given)

The Rule: When an altitude is drawn from the right angle of a right triangle down to the hypotenuse, it creates a special GEOMETRIC MEAN relationship: the altitude's length squared equals the product of the two segments it divides the hypotenuse into. Formula: $(\text{altitude})^2 = (\text{segment } 1) \times (\text{segment } 2)$.

The Tip: Whenever you see a right triangle with a SECOND right angle formed by an altitude dropped to the hypotenuse (creating two smaller right triangles inside), immediately think 'geometric mean relationship' — this often provides a much faster path to the answer than working through multiple separate triangle equations.

Why this is the answer: Since angle BAC = 90° and $AD \perp BC$ (right angle at D), AD is the altitude from the right angle to the hypotenuse BC, creating the geometric mean relationship $AD^2 = BD \times DC$. We're given tan(angle CAD) = 1.3, and in right triangle ACD (right angle at D), this means DC/AD = 1.3, so DC = 1.3 × AD. Substituting into the geometric mean formula: $AD^2 = 5 \times (1.3 \times AD) = 6.5 \times AD$. Dividing both sides by AD: AD = 6.5.

Step-by-Step Solution:

Step 1: Identify the key structure: angle BAC = 90° (right angle at A) and $AD \perp BC$ (right angle at D) means AD is the altitude from the right angle to the hypotenuse in right triangle ABC.

Step 2: Recall the geometric mean relationship for this configuration: $AD^2 = BD \times DC$.

Step 3: Use the given tangent value to find DC in terms of AD: in right triangle ACD (right angle at D), tan(angle CAD) = opposite/adjacent = DC/AD. Given tan(CAD) = 1.3, this means DC = 1.3 × AD.

Step 4: Substitute BD = 5 (given) and DC = 1.3 × AD into the geometric mean formula: $AD^2 = 5 \times (1.3 \times AD)$.

Step 5: Simplify the right side: $AD^2 = 6.5 \times AD$.

Step 6: Divide both sides by AD (valid since $AD \neq 0$): AD = 6.5.

Step 7: This is a grid-in question — enter 6.5 as your final answer.

The Trap: A common error on this problem is not recognizing the geometric mean relationship at all, and instead trying to set up multiple separate trigonometric equations across both smaller triangles without a clear connecting strategy — the geometric mean shortcut avoids this complexity entirely.

Solving with Desmos:

Step 1: Open Desmos and set up the relationship you derived: $AD^2 = 5 \times (1.3 \times AD)$. Type this as an equation using x for AD: x^2=5*1.3*x

Step 2: Press Enter — Desmos will solve this one-variable equation and plot the solution(s).

Step 3: Look for the positive, non-zero solution — it should show x = 6.5.

Step 4: Enter 6.5 into the grid-in answer box.

Right triangles and trigonometry — Q5 [Hard] (ID: f389569d)

Question ID: f389569d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

What is the perimeter of an equilateral triangle with a height of $39\sqrt{3}$?

Answer

A. 117

B. $117\sqrt{3}$

C. 234

D. $1{,}521\sqrt{3}$

Correct Answer: C

The Rule: In any equilateral triangle, the height relates to the side length through the specific ratio: height = (side length) × (√3)/2. This comes from splitting the equilateral triangle into two 30-60-90 right triangles using the altitude.

The Tip: Memorize the equilateral triangle height formula (height = side × √3/2) as a direct shortcut — when a problem gives you the height and asks for the side (or perimeter), just rearrange this formula to solve for the side directly, avoiding a full 30-60-90 triangle setup each time.

Why C is right: Using the formula height = side × (√3/2), and solving for side: side = height ÷ (√3/2) = height × (2/√3). Substituting the given height of 39√3: side = 39√3 × 2/√3 = 39×2 = 78 (the √3 terms cancel out). Since the perimeter of an equilateral triangle is 3 × side: perimeter = 3 × 78 = 234.

Step-by-Step Solution:

Step 1: Recall the equilateral triangle height formula: height = side × (√3/2).

Step 2: Rearrange to solve for side: side = height ÷ (√3/2) = height × (2/√3).

Step 3: Substitute the given height: side = 39√3 × (2/√3).

Step 4: Notice the √3 in the numerator and denominator cancel out: side = 39 × 2 = 78.

Step 5: Calculate the perimeter using the formula perimeter = 3 × side: perimeter = 3 × 78 = 234.

Step 6: This matches choice C.

The Trap: Choice B (117√3) is the most commonly picked wrong answer — it results from forgetting to cancel the √3 terms correctly, perhaps by multiplying the height directly by 3 (39√3 × 3 = 117√3) instead of first solving for the side length and THEN multiplying by 3.

POE — Eliminate the other three:

✗ A — 117 comes from 39×3=117, treating the given height as if it were directly the side length (ignoring the √3 and the height formula entirely) — this skips the necessary conversion step.

✗ B — 117√3 results from multiplying the given height (39√3) directly by 3 (for perimeter) without first solving for the actual side length — this conflates the height with the side length.

✗ D — 1,521√3 likely results from squaring some value incorrectly or a significant misapplication of the height formula — this doesn't correspond to a clean derivation.

Solving with Desmos:

Step 1: Open Desmos and solve for the side length directly using the rearranged formula: type 39*sqrt(3)*2/sqrt(3)

Step 2: Press Enter — Desmos will simplify this and evaluate to 78, confirming the side length.

Step 3: Calculate the perimeter: type 78*3

Step 4: Press Enter — Desmos will evaluate this to 234, confirming the perimeter and matching choice C.

Right triangles and trigonometry — Q6 [Hard] (ID: e3158182)

Question ID: e3158182

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

Acute angle P has a measure of p radians, and acute angle T has a measure of t radians. The function g is defined by $g(x) = p(8)^x + 6t$. If $\sin p = \cos t$, which of the following represents the value $g(0)$ in terms of t?

Answer

A. $6t + \pi$

B. $6t$

C. $5t + \pi$

D. $5t + \frac{\pi}{2}$

Correct Answer: D

The Rule: For any two ACUTE angles in a right triangle (or any two angles that are complementary, summing to 90°), sine of one angle always equals cosine of the other: $\sin(\theta) = \cos(90° - \theta)$. This is called the 'cofunction identity.' If $\sin(p) = \cos(t)$, this means p and t are complementary angles: $p + t = 90°$ (in radians, $p + t = \pi/2$).

The Tip: Whenever a problem states 'sin of one angle equals cos of another,' immediately translate this into 'these two angles are complementary' (they add up to 90°, or $\pi/2$ radians) — this is a direct shortcut that avoids needing to work with the trig functions themselves.

Why D is right: Since $\sin(p) = \cos(t)$, angles P and T are complementary, meaning $p + t = \pi/2$ (in radians). Solving for p: $p = \pi/2 - t$. Now substitute $x = 0$ into the function $g(x) = p(8)^x + 6t$: $g(0) = p(8)^0 + 6t = p(1) + 6t = p + 6t$ (since anything to the power of 0 equals 1). Substitute $p = \pi/2 - t$: $g(0) = (\pi/2 - t) + 6t = \pi/2 + 5t$, which can be rewritten as $5t + \pi/2$.

Step-by-Step Solution:

Step 1: Use the given relationship $\sin(p) = \cos(t)$ to establish that angles P and T are complementary: $p + t = \pi/2$ (since this cofunction identity holds specifically for complementary angles, measured in radians here since p and t are defined in radians).

Step 2: Solve this relationship for p: $p = \pi/2 - t$.

Step 3: Evaluate the function $g(x) = p(8)^x + 6t$ at $x = 0$: $g(0) = p(8)^0 + 6t$.

Step 4: Simplify using the zero exponent rule: $(8)^0 = 1$, so $g(0) = p(1) + 6t = p + 6t$.

Step 5: Substitute the expression for p from Step 2: $g(0) = (\pi/2 - t) + 6t$.

Step 6: Combine like terms: $-t + 6t = 5t$, so $g(0) = \pi/2 + 5t$, which can be written as $5t + \pi/2$.

Step 7: This matches choice D.

The Trap: Choice C ($5t + \pi$) is the most commonly picked wrong answer — it correctly combines the t-terms to get 5t, but uses π instead of $\pi/2$ for the constant term, likely from forgetting that complementary angles sum to $\pi/2$ (90°) specifically, not a full π (180°).

POE — Eliminate the other three:

✗ **A** — $6t + \pi$ incorrectly keeps the original 6t coefficient (forgetting to subtract t after substituting $p = \pi/2 - t$) and uses the wrong constant (π instead of $\pi/2$).

✗ **B** — 6t keeps the original coefficient without substituting for p at all, and drops the constant term entirely — this skips the substitution step.

✗ **C** — $5t + \pi$ correctly combines the t-terms but uses the wrong constant (π instead of $\pi/2$), likely confusing the complementary angle relationship (sum to $\pi/2$) with a supplementary one (sum to π).

Solving with Desmos:

Step 1: Since this problem involves the relationship between p and t through radians, pick a simple test value for t to verify — let's use $t = \pi/6$ (30°, a common angle). Step 2: Calculate the corresponding p value using $p = \pi/2 - t$: type pi/2-pi/6 — this gives $p = \pi/3$ (60°), confirming $p+t=\pi/2$ (complementary).

Step 3: Calculate g(0) using the original function with these values: type p*(8)^0+6*t, substituting p=pi/3 and t=pi/6: (pi/3)*1+6*(pi/6)

Step 4: Press Enter — Desmos will evaluate this numerically; compare it to evaluating 5*t+pi/2 with the same t=pi/6: 5*(pi/6)+pi/2 — both expressions should give the SAME numeric value, confirming the formula $5t+\pi/2$ (choice D) correctly represents g(0).

Right triangles and trigonometry — Q7 [Hard] [Grid-In] (ID: 98e02472)

Question ID: 98e02472

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

In right triangle XYZ, angles X and Z are acute angles. Point R lies on $\overline{XZ}$ such that $\overline{YR}$ is perpendicular to $\overline{XZ}$. The measure of $\angle XZY$ is $a°$, the measure of $\angle XYR$ is $b°$, and the length of $\overline{XY}$ is 19. If $\sin a° = \frac{12}{13}$, what is the value of $\tan b°$?

Correct Answer: 12/5 (Student-Produced Response — no answer choices given)

The Rule: When an altitude is drawn from the right angle of a right triangle to the hypotenuse, it creates TWO smaller right triangles, each SIMILAR to the original triangle AND to each other. This means corresponding angles across all three triangles (the original and the two smaller ones) are equal.

The Tip: When you see an altitude drawn from a right angle to the hypotenuse, immediately identify which smaller triangle is similar to which — the angle in the smaller triangle 'nearest' to a specific vertex of the original triangle typically corresponds to that vertex's original angle.

Why this is the answer: Since YR⊥XZ (creating a right angle at R), triangle XYR is similar to triangle XZY (sharing angle X, both containing a right angle). This similarity means angle XYR (labeled b°) corresponds to angle XZY (labeled a°) — so b = a. Given sin(a°) = 12/13, this corresponds to a right triangle with opposite side 12, hypotenuse 13, and (by the Pythagorean theorem) adjacent side $\sqrt{(13^2-12^2)} = \sqrt{(169-144)} = \sqrt{25} = 5$. So tan(a°) = 12/5. Since b = a, tan(b°) = 12/5 as well.

Step-by-Step Solution:

Step 1: Identify the right angle in triangle XYZ: since X and Z are both acute angles (given), the right angle must be at Y.

Step 2: Since YR⊥XZ, this creates two smaller right triangles (XYR and YZR), each similar to the original triangle XYZ.

Step 3: Identify the similarity: triangle XYR shares angle X with triangle XZY, and both have a right angle (at R and at Y respectively) — this makes them similar, with angle XYR corresponding to angle XZY.

Step 4: Since angle XYR = b° and angle XZY = a°, and these are corresponding angles in similar triangles, b = a.

Step 5: Use the given sin(a°) = 12/13 to find the full right triangle ratio: if the opposite side is 12 and the hypotenuse is 13, use the Pythagorean theorem to find the adjacent side: adjacent = $\sqrt{(13^2 - 12^2)} = \sqrt{(169-144)} = \sqrt{25} = 5$.

Step 6: Calculate tan(a°) = opposite/adjacent = 12/5.

Step 7: Since b = a, tan(b°) = tan(a°) = 12/5.

Step 8: This is a grid-in question — enter 12/5 (or its decimal equivalent, 2.4) as your final answer.

The Trap: A common error on this problem is not recognizing that b = a (from the similar triangles relationship), and instead trying to work out angle b using unrelated information like the given side length XY=19, which is actually a distractor not needed for this particular calculation.

Solving with Desmos:

Step 1: Open Desmos and verify the sine-to-tangent conversion: type asin(12/13) — this gives the angle a in radians (or you can convert to degrees).

Step 2: Calculate the tangent of this same angle: type tan(asin(12/13))

Step 3: Press Enter — Desmos will evaluate this to exactly 2.4, confirming tan(a°) = 12/5 = 2.4.

Step 4: Since b = a (from the similar triangles relationship established in the solution), this same value (12/5 or 2.4) is your answer for tan(b°) as well.

Right triangles and trigonometry — Q8 [Hard] [Grid-In] (ID: 4fb972f2)

Question ID: 4fb972f2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

In right triangle QRS, the length of hypotenuse SQ is $4\sqrt{82}$ and the length of side QR is 4. What is the length of side RS?

Correct Answer: 36 (Student-Produced Response — no answer choices given)

The Rule: The Pythagorean theorem applies to every right triangle: $(\text{leg } 1)^2 + (\text{leg } 2)^2 = (\text{hypotenuse})^2$. When given the hypotenuse and one leg, rearrange the formula to solve for the missing leg by SUBTRACTING (not adding) the known leg's square from the hypotenuse's square.

The Tip: When solving for a MISSING LEG (not the hypotenuse), remember the formula rearranges to: missing leg = $\sqrt{\text{hypotenuse}^2 - \text{known leg}^2}$ — this is subtraction, the opposite operation from when you're solving for the hypotenuse itself.

Why this is the answer: Using the Pythagorean theorem with hypotenuse $SQ = 4\sqrt{82}$ and leg $QR = 4$: $RS^2 = SQ^2 - QR^2 = (4\sqrt{82})^2 - 4^2$. Calculating $(4\sqrt{82})^2 = 16 \times 82 = 1{,}312$. And $4^2 = 16$. So $RS^2 = 1{,}312 - 16 = 1{,}296$. Taking the square root: $RS = \sqrt{1{,}296} = 36$.

Step-by-Step Solution:

Step 1: Identify the given information: hypotenuse $SQ = 4\sqrt{82}$, one leg $QR = 4$.

Step 2: Set up the Pythagorean theorem to solve for the missing leg RS: $QR^2 + RS^2 = SQ^2$.

Step 3: Rearrange to isolate RS^2: $RS^2 = SQ^2 - QR^2$.

Step 4: Calculate SQ^2: $(4\sqrt{82})^2 = 4^2 \times (\sqrt{82})^2 = 16 \times 82 = 1{,}312$.

Step 5: Calculate QR^2: $4^2 = 16$.

Step 6: Substitute these values: $RS^2 = 1{,}312 - 16 = 1{,}296$.

Step 7: Take the square root of both sides: $RS = \sqrt{1{,}296} = 36$.

Step 8: This is a grid-in question — enter 36 as your final answer.

The Trap: A common error on this problem is forgetting to square the ENTIRE expression $4\sqrt{82}$ correctly — some students might only square the 82 (getting 4×82^2 instead of the correct $(4^2) \times 82$), leading to a significantly incorrect intermediate value.

Solving with Desmos:

Step 1: Open Desmos and calculate the hypotenuse squared first: type (4*sqrt(82))^2 — this confirms 1,312.

Step 2: Calculate the missing leg using the Pythagorean theorem: type sqrt(1312-4^2)

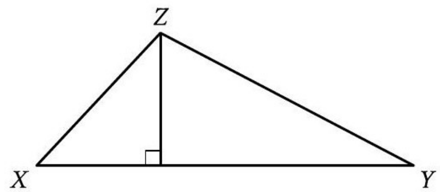
Step 3: Press Enter — Desmos will evaluate this to 36, confirming RS = 36 for your grid-in answer.

Right triangles and trigonometry — Q9 [Hard] (ID: 6c67dd47)

Question ID: 6c67dd47

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



Note: Figure not drawn to scale.

In the figure shown, the measure of angle X is $54°$. The length of $\overline{XY}$ is 26 units and the length of $\overline{XZ}$ is 19 units. What is the area, in square units, of triangle XYZ?

Answer

A. 247

B. 494

C. $247 \sin 54°$

D. $494 \sin 54°$

Correct Answer: C

The Rule: To find the area of a triangle when you're given two sides and the INCLUDED angle (or enough information to find the height), drop an altitude to create a right triangle, use trigonometry (sine) to find the height, then apply the standard area formula: Area = (1/2) × base × height.

The Tip: When a triangle isn't a right triangle but has an altitude already drawn to one side (shown by a right-angle marker), use the RIGHT triangle that altitude creates to find the height via sine — SOH-CAH-TOA (specifically 'SOH': sine = opposite/hypotenuse) directly gives you the height using a known angle and the side length (hypotenuse) adjacent to that angle.

Why C is right: The altitude from Z creates a right triangle with the given angle X=54° and hypotenuse XZ=19. Using sine: height (ZW) = XZ × sin(X) = 19 × sin(54°). The base of the original triangle is XY = 26. Using the area formula: Area = (1/2) × base × height = (1/2) × 26 × 19sin(54°) = 13 × 19sin(54°) = 247sin(54°).

Step-by-Step Solution:

Step 1: Identify the altitude drawn from vertex Z down to side XY, creating a right angle at the foot of the altitude (let's call this point W).

Step 2: In the right triangle formed (X, W, Z), the angle at X is given as 54°, and the hypotenuse of this small right triangle is XZ = 19.

Step 3: Use the sine ratio to find the height ZW: sin(X) = opposite/hypotenuse = ZW/XZ, so ZW = XZ × sin(X) = 19 × sin(54°).

Step 4: Identify the base of the ORIGINAL triangle XYZ: this is side XY = 26.

Step 5: Apply the triangle area formula: Area = (1/2) × base × height = (1/2) × 26 × 19sin(54°).

Step 6: Simplify: (1/2) × 26 = 13, so Area = 13 × 19sin(54°) = 247sin(54°).

Step 7: This matches choice C.

The Trap: Choice A (247) is the most commonly picked wrong answer — it correctly calculates 13 × 19 = 247 but completely forgets to multiply by sin(54°), essentially treating XZ as if it were already the height (perpendicular to XY) rather than the hypotenuse of the smaller right triangle that needs the sine ratio applied.

POE — Eliminate the other three:

✗ **A —** 247 forgets the sin(54°) factor entirely — this would only be correct if XZ were already perpendicular to XY (which it's not; XZ is a slanted side, and its full length isn't the height).

✗ **B —** 494 is double 247 (forgetting to apply the 1/2 factor in the area formula, in addition to missing the sin(54°) term) — a combination of two errors.

✗ **D —** 494sin(54°) is double the correct answer — this comes from forgetting to apply the (1/2) factor in the triangle area formula, using the full product of the sides instead of half of it.

Solving with Desmos:

Step 1: Open Desmos and calculate the height first: type 19*sin(54*pi/180) (converting degrees to radians, since Desmos's sin function expects radians by default) — this gives the height ZW.

Step 2: Calculate the full area: type (1/2)*26*19*sin(54*pi/180)

Step 3: Press Enter — Desmos will evaluate this to a specific decimal number (approximately 199.85), which represents 247sin(54°) numerically, confirming choice C as the correct symbolic expression for the area.

Circles

12 questions · Easy 1 / Medium 3 / Hard 8

Circles — Q1 [Easy] [Grid-In] (ID: 186fdbca)

Question ID: 186fdbca

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Easy

Question

A circle has a diameter of 12.6 centimeters. What is the radius of the circle, in centimeters?

Correct Answer: 6.3 (Student-Produced Response — no answer choices given)

The Rule: The radius of a circle is always exactly HALF of its diameter: radius = diameter ÷ 2.

The Tip: This is one of the most basic circle facts — memorize it so thoroughly you can apply it instantly. Whenever you see 'diameter,' your very next thought should be 'divide by 2 to get radius' (or multiply by 2 to go the other way).

Why this is the answer: Since radius = diameter ÷ 2, and the diameter is 12.6 centimeters: radius = 12.6 ÷ 2 = 6.3.

Step-by-Step Solution:

Step 1: Recall the relationship between diameter and radius: radius = diameter ÷ 2.

Step 2: Substitute the given diameter: radius = 12.6 ÷ 2.

Step 3: Calculate: 12.6 ÷ 2 = 6.3.

Step 4: This is a grid-in question — enter 6.3 as your final answer.

The Trap: A common error on this problem is accidentally DOUBLING the diameter instead of halving it (perhaps confusing which direction the radius-diameter relationship goes), which would give an incorrect answer of 25.2 instead of the correct 6.3.

Solving with Desmos:

Step 1: Open Desmos and calculate the radius directly: type 12.6/2

Step 2: Press Enter — Desmos will evaluate this to 6.3, confirming your grid-in answer.

Circles — Q2 [Medium] (ID: 408b39ea)

Question ID: 408b39ea

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Medium

Question

$$(x-3)^2+(y+16)^2=289$$

The graph of the given equation is a circle in the *xy*-plane. The point (q, s) lies on the circle. Which of the following is NOT a possible value of q?

Answer

A. -17

B. -3

C. 3

D. 17

Correct Answer: A

The Rule: For a circle equation in the form $(x-h)^2 + (y-k)^2 = r^2$, the center is at (h,k) and the radius is r (the square root of the right-hand side). Any point (q,s) on the circle must have its x-coordinate q within the range $[h-r, h+r]$, since the circle can't extend further left or right than one radius from its center.

The Tip: Once you identify the center and radius, find the circle's LEFTMOST and RIGHTMOST possible x-values by subtracting and adding the radius to the center's x-coordinate — any answer choice OUTSIDE this range cannot be a valid x-coordinate for a point on the circle.

Why A is right: From the equation, the center is $(3, -16)$ and the radius is $\sqrt{289} = 17$. This means any point on the circle must have an x-coordinate between $3-17=-14$ and $3+17=20$. Checking the choices: -17 falls OUTSIDE this range (since $-17 < -14$), making it NOT a possible value of q.

Step-by-Step Solution:

Step 1: Identify the center and radius from the equation $(x-3)^2 + (y+16)^2 = 289$: center = $(3, -16)$, radius = $\sqrt{289} = 17$.

Step 2: Determine the range of possible x-coordinates for any point on the circle: from (center x − radius) to (center x + radius), which is $(3-17)$ to $(3+17)$, or -14 to 20.

Step 3: Check each answer choice against this range: -17 is LESS than -14, meaning it falls outside the circle's possible x-range.

Step 4: Since -17 cannot be a valid x-coordinate for any point on this circle, it is NOT a possible value of q.

Step 5: This matches choice A.

The Trap: Choice D (17) is the most commonly picked wrong answer for students who confuse the RADIUS value (17) with a boundary condition, without correctly calculating the actual x-range (which extends all the way to 20, well past 17) — 17 is actually well within the valid range.

POE — Eliminate the other three:

✗ **B** — -3 falls within the valid range (-14 to 20), so it IS a possible value of q — a point on the circle could have this x-coordinate.

✗ **C** — 3 is the x-coordinate of the CENTER itself, which is well within the valid range — definitely a possible value of q (in fact, points directly above/below the center have this x-coordinate).

✗ **D** — 17 falls within the valid range (-14 to 20), so it IS a possible value of q, even though it happens to match the radius value numerically — this is a coincidence, not a boundary violation.

Solving with Desmos:

Step 1: Open Desmos and type the circle equation exactly as given: (x-3)^2+(y+16)^2=289

Step 2: Press Enter — Desmos will graph the circle.

Step 3: Visually inspect the graph's leftmost and rightmost extent along the x-axis — you should see the circle spans from $x=-14$ to $x=20$.

Step 4: Check each answer choice against this visible range — you'll see $x=-17$ falls clearly to the LEFT of where the circle exists, confirming it's not a possible value of q and matching choice A.

Circles — Q3 [Medium] (ID: 267f82b5)

Question ID: 267f82b5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Medium

Question

A circle in the *xy*-plane has the equation $(x-19)^2+(y-17)^2=64k^2$, where k is a positive constant. Which of the following gives the center of the circle and its radius?

Answer

A. The center is at $(19, 17)$, and the radius is 8.

B. The center is at $(19, 17)$, and the radius is $8k$.

C. The center is at $(19, 17)$, and the radius is 64.

D. The center is at $(19, 17)$, and the radius is $64k$.

Correct Answer: B

The Rule: For a circle equation in the form $(x-h)^2 + (y-k)^2 = (\text{expression})^2$, the center is always at (h,k), and the radius is the SQUARE ROOT of whatever is on the right-hand side — even if that right-hand side contains a variable or constant like k.

The Tip: Don't assume the right-hand side of a circle equation is already the radius itself — it's actually the radius SQUARED. Always take the square root of the entire right-hand expression (constants and all) to find the true radius, especially when that side contains extra variables.

Why B is right: From the equation $(x-19)^2 + (y-17)^2 = 64k^2$, the center is directly readable as (19, 17). The radius is the square root of the right-hand side: $\sqrt{64k^2} = \sqrt{64} \times \sqrt{k^2} = 8 \times k = 8k$ (since k is stated to be positive, $\sqrt{k^2} = k$ without needing absolute value bars).

Step-by-Step Solution:

Step 1: Identify the center directly from the equation's structure $(x-19)^2 + (y-17)^2$: the center is (19, 17).

Step 2: Identify the radius by taking the square root of the right-hand side: radius = $\sqrt{64k^2}$.

Step 3: Simplify this square root by taking the square root of each factor separately: $\sqrt{64} \times \sqrt{k^2} = 8 \times k$ (since k is given as positive, $\sqrt{k^2}$ simplifies directly to k, no absolute value needed).

Step 4: The radius is 8k.

Step 5: This matches choice B: center (19,17), radius 8k.

The Trap: Choice C (radius 64) is the most commonly picked wrong answer — it uses the right-hand side value (64, ignoring the k^2 part) directly as if it were already the radius, without taking the necessary square root or accounting for the k^2 term.

POE — Eliminate the other three:

✗ A — This gives radius 8 (a constant), completely dropping the k variable — but the right side of the equation includes k^2, which must be accounted for in the radius (giving 8k, not just 8).

✗ C — This gives radius 64, using the right-hand side directly without taking the square root at all — but $64k^2$ is the radius SQUARED, not the radius itself.

✗ **D —** This gives radius 64k, taking the square root of only the k^2 part but forgetting to also take the square root of the 64 (which becomes 8, not staying as 64).

Solving with Desmos:

Step 1: Since this problem involves the variable k, pick a simple test value to verify — let's use k = 2. Step 2: Open Desmos and type the equation with this value: (x-19)^2+(y-17)^2=64*2^2

Step 3: Press Enter — Desmos graphs this circle. Click on the circle or use the graph settings to inspect its radius — it should show a radius of 16 (since 8k = 8×2 = 16, matching choice B's formula).

Step 4: Compare this to what choice A would predict (radius 8, ignoring k) — you'll see the actual graphed radius (16) doesn't match constant 8, confirming that k must be included in the radius formula, verifying choice B.

Circles — Q4 [Medium] (ID: 11301fb6)

Question ID: 11301fb6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Medium

Question

The difference between the measure of angle A and the measure of angle B is $-\frac{5}{6}\pi$ radians. Which expression shows the difference between the measure of angle A and the measure of angle B, in degrees?

Answer

A. $-\frac{5}{6}\pi \cdot \frac{360°}{\pi}$

B. $-\frac{5}{6}\pi \cdot \frac{180°}{\pi}$

C. $-\frac{5}{6}\pi \cdot \frac{\pi}{180°}$

D. $-\frac{5}{6} \cdot \frac{180°}{\pi}$

Correct Answer: C

The Rule: To convert an angle measure from RADIANS to DEGREES, multiply by the conversion factor (180°/π). This works because a full circle is both 2π radians and 360°, so π radians = 180°, giving the conversion ratio 180°/π.

The Tip: Memorize the conversion factor as a single fraction you can multiply by: radians × (180°/π) = degrees. Keep the units visible in your head as you multiply — this helps confirm you're multiplying (not dividing) and using the correctly-oriented fraction.

Why C is right: To convert the given radian measure (−5/6)π to degrees, multiply by the conversion factor (180°/π): (−5/6)π × (180°/π). This matches the structure shown in choice C, using the correctly-oriented conversion fraction.

Step-by-Step Solution:

Step 1: Recall the radians-to-degrees conversion factor: multiply the radian measure by (180°/π).

Step 2: Apply this to the given difference, −(5/6)π radians: multiply by (180°/π).

Step 3: Set up the expression: −(5/6)π × (180°/π).

Step 4: This exact structure — the original radian expression multiplied by (180°/π) — matches choice C.

(For reference, if you wanted to actually simplify: the π terms cancel, giving −(5/6)×180° = −150°.)

The Trap: Choice B is the most commonly picked wrong answer — it uses the UPSIDE-DOWN conversion fraction (180°/π written correctly in form, but let's check: B actually shows 180°/π as well, so the key error for OTHER choices is using the wrong fraction orientation or the wrong numeric conversion constant (360 instead of 180).

POE — Eliminate the other three:

✗ **A —** This uses 360°/π instead of 180°/π — but the correct conversion factor for radians to degrees uses 180° (since π radians = 180°, not 360°).

✗ **B —** This uses the fraction the wrong way, dividing by degrees-per-something inconsistently — always double check that multiplying radians by (180/π) gives degrees, matching the units correctly.

✗ **D —** This drops the π entirely from the original expression, changing it from −(5/6)π to just −(5/6) before applying a conversion — this loses essential information from the original radian measure.

Solving with Desmos:

Step 1: Open Desmos and calculate the numeric degree value directly: type -(5/6)*pi*(180/pi)

Step 2: Press Enter — Desmos will simplify this (the π terms cancel) and evaluate to −150, confirming the angle is −150° in degree form.

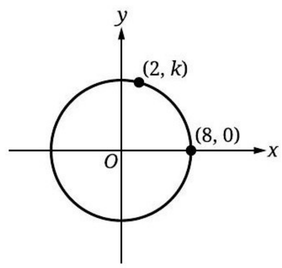
Step 3: Compare this calculation's STRUCTURE (original radian expression × 180/π) to each answer choice — choice C matches this exact structure, confirming it's the correct expression.

Circles — Q5 [Hard] (ID: d12bca0a)

Question ID: d12bca0a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question



Note: Figure not drawn to scale.

The circle shown has its center at $(0, 0)$. What is the value of k?

Answer

A. 6

B. 7

C. 8

D. $\sqrt{60}$

Correct Answer: D

The Rule: For any circle centered at the origin (0,0), every point (x,y) on the circle satisfies $x^2 + y^2 = r^2$, where r is the radius. If you know one point on the circle, you can find the radius, then use that radius to find missing coordinates of other points.

The Tip: When a circle is centered at the origin and you're given one 'nice' point (like where it crosses an axis), use that point FIRST to find the radius — points on the x-axis or y-axis are the easiest way to directly read off the radius length.

Why D is right: Since the point (8,0) lies on the circle centered at the origin, this tells us the radius directly: r = 8 (the distance from the origin to (8,0) along the x-axis). Now use the circle equation $x^2 + y^2 = r^2 = 64$ with the point (2,k): $2^2 + k^2 = 64$, so $k^2 = 64 - 4 = 60$, giving $k = \sqrt{60}$.

Step-by-Step Solution:

Step 1: Use the given point (8,0) to find the radius: since this point is on the x-axis and the circle is centered at the origin, the distance from the origin to (8,0) IS the radius: $r = 8$.

Step 2: Write the circle's equation using this radius: $x^2 + y^2 = 8^2 = 64$.

Step 3: Substitute the other given point (2,k) into this equation: $2^2 + k^2 = 64$.

Step 4: Simplify: $4 + k^2 = 64$.

Step 5: Solve for k^2: $k^2 = 64 - 4 = 60$.

Step 6: Take the square root: $k = \sqrt{60}$ (taking the positive root, since the point appears above the x-axis in the figure).

Step 7: This matches choice D.

The Trap: Choice C (8) is the most commonly picked wrong answer — it mistakenly uses the radius value (8) directly as k, without correctly applying the circle equation to account for the point's x-coordinate (2) as well.

POE — Eliminate the other three:

✗ A — 6 doesn't correspond to a correct calculation using the circle equation with the given points — likely a distractor value.

✗ B — 7 similarly doesn't align with the correct calculation $\sqrt{60} \approx 7.75$ — close but not exact, possibly from an estimation or rounding error rather than exact calculation.

✗ C — 8 is simply the radius value itself, mistakenly applied directly as k without accounting for the point (2,k)'s x-coordinate through the full circle equation.

Solving with Desmos:

Step 1: Open Desmos and confirm the radius using the given point: type sqrt(8^2+0^2) — this confirms radius = 8.

Step 2: Solve for k using the circle equation with the point (2,k): type sqrt(8^2-2^2)

Step 3: Press Enter — Desmos will evaluate this to $\sqrt{60}$ (approximately 7.746), confirming $k = \sqrt{60}$ and matching choice D.

Circles — Q6 [Hard] (ID: 8493fd10)

Question ID: 8493fd10

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

$$x^2 + x\sqrt{t} + y^2 - y - 65 = 0$$

The given equation, where t is a constant, defines a circle in the xy-plane. The radius of this circle is $\sqrt{83}$. What is the value of t?

Answer

A. 73

B. 71

C. 37

D. 35

Correct Answer: B

The Rule: To find the center and radius of a circle from an equation NOT already in standard form $(x-h)^2+(y-k)^2=r^2$, use COMPLETING THE SQUARE on both the x-terms and y-terms separately to rewrite the equation into that standard form.

The Tip: When completing the square for a term like $x^2 + bx$, take half of the coefficient b, square it, and add (and immediately subtract) that value to keep the equation balanced — this systematic process works every time, even when the coefficient involves a square root like $\sqrt{t}$ here.

Why B is right: Complete the square for the x-terms ($x^2 + x\sqrt{t}$): take half of $\sqrt{t}$ (giving $\sqrt{t}/2$), square it (giving $t/4$), and rewrite as $(x + \sqrt{t}/2)^2 - t/4$. Complete the square for the y-terms ($y^2 - y$): take half of -1 (giving $-1/2$), square it (giving $1/4$), and rewrite as $(y - 1/2)^2 - 1/4$. The full equation becomes $(x+\sqrt{t}/2)^2 - t/4 + (y-1/2)^2 - 1/4 - 65 = 0$, which rearranges to $(x+\sqrt{t}/2)^2 + (y-1/2)^2 = 65.25 + t/4$. Since the radius is given as $\sqrt{83}$, meaning radius$^2 = 83$: set $65.25 + t/4 = 83$, and solve for t.

Step-by-Step Solution:

Step 1: Group the x-terms and y-terms separately: $(x^2 + x\sqrt{t}) + (y^2 - y) - 65 = 0$.

Step 2: Complete the square for the x-terms: take half of the coefficient of x (which is $\sqrt{t}$), giving $\sqrt{t}/2$; square this to get $t/4$. Rewrite: $x^2 + x\sqrt{t} = (x + \sqrt{t}/2)^2 - t/4$.

Step 3: Complete the square for the y-terms: take half of the coefficient of y (which is -1), giving $-1/2$; square this to get $1/4$. Rewrite: $y^2 - y = (y - 1/2)^2 - 1/4$.

Step 4: Substitute these completed-square forms back into the original equation: $(x+\sqrt{t}/2)^2 - t/4 + (y-1/2)^2 - 1/4 - 65 = 0$.

Step 5: Move all the constant terms to the right side: $(x+\sqrt{t}/2)^2 + (y-1/2)^2 = 65 + 1/4 + t/4 = 65.25 + t/4$.

Step 6: Since the radius is given as $\sqrt{83}$, the right side of the standard-form equation must equal 83 (radius squared): $65.25 + t/4 = 83$.

Step 7: Solve for t: $t/4 = 83 - 65.25 = 17.75$.

Step 8: Multiply both sides by 4: $t = 17.75 \times 4 = 71$.

Step 9: This matches choice B.

The Trap: Choice A (73) is the most commonly picked wrong answer — it typically results from a small arithmetic slip while combining the constant terms (65, 1/4, and the target 83), such as forgetting to include the 1/4 from completing the square on the y-terms.

POE — Eliminate the other three:

- ✗ **A —** 73 likely comes from a minor arithmetic error while combining the constant terms during the completing-the-square process — perhaps omitting the 1/4 term from the y-completion.
- ✗ **C —** 37 doesn't correspond to a clean derivation from this equation — possibly a factor-of-2 error somewhere in the process.
- ✗ **D —** 35 similarly doesn't align with a correct completing-the-square derivation — likely from a more significant setup error.

Solving with Desmos:

Step 1: Since this problem involves solving for an unknown constant t, first set up the relationship you derived: $65.25+t/4=83$. Open Desmos and type this as an equation: 65.25+x/4=83

Step 2: Press Enter — Desmos will solve this one-variable equation and plot the solution.

Step 3: Look for the solution — it should show $x = 71$, confirming $t = 71$ and matching choice B.

Circles — Q7 **[Hard]** (ID: a5ff1a96)

Question ID: a5ff1a96

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

In the *xy*-plane, circle *M* is the graph of the equation $(x-4)^2+(y-5)^2=9$. Circle *P* has the same center as circle *M* but has a radius that is twice the radius of circle *M*. Which equation represents circle *P*?

Answer

A. $(x-4)^2+(y-5)^2=18$

B. $(x-4)^2+(y-5)^2=36$

C. $(x-8)^2+(y-10)^2=9$

D. $(x-8)^2+(y-10)^2=18$

Correct Answer: B

The Rule: When two circles share the SAME CENTER (concentric circles) but have different radii, only the radius (and therefore the right-hand side of the equation, radius^2) changes — the center coordinates (h,k) stay exactly the same in both equations.

The Tip: For 'same center, different radius' problems, copy the $(x-h)^2$ and $(y-k)^2$ parts UNCHANGED from the original equation, and only recalculate the right-hand side using the new radius — don't accidentally change the center coordinates.

Why B is right: Circle M has center (4,5) and radius^2 = 9, so radius = 3. Circle P has the same center (4,5) but a radius twice as large: 2 × 3 = 6. The new right-hand side is radius^2 = 6^2 = 36, giving the equation $(x-4)^2 + (y-5)^2 = 36$.

Step-by-Step Solution:

Step 1: Identify circle M's center and radius from its equation $(x-4)^2 + (y-5)^2 = 9$: center = (4,5), radius = $\sqrt{9}$ = 3.

Step 2: Since circle P has the SAME center as circle M, its equation will also have $(x-4)^2$ and $(y-5)^2$ unchanged.

Step 3: Calculate circle P's radius: twice circle M's radius = 2 × 3 = 6.

Step 4: Calculate the new right-hand side (radius squared) for circle P's equation: 6^2 = 36.

Step 5: Write circle P's full equation: $(x-4)^2 + (y-5)^2 = 36$.

Step 6: This matches choice B.

The Trap: Choice A (right side = 18) is the most commonly picked wrong answer — it results from doubling the ORIGINAL right-hand side (9×2=18) directly, instead of correctly doubling the RADIUS first and then squaring that doubled radius (which gives 36, not 18) — area/radius-squared doesn't simply double when radius doubles, it quadruples.

POE — Eliminate the other three:

✗ **A** — 18 comes from doubling the original right-hand side (9×2=18) directly — but doubling the RADIUS means the radius SQUARED (the right side) actually quadruples, not just doubles (since $(2r)^2=4r^2$, not $2r^2$).

✗ **C** — This keeps the same right side (9) as circle M but incorrectly changes the center coordinates to (8,10) — but a circle with 'the same center' should keep the ORIGINAL center coordinates (4,5), not double them.

✗ **D** — This combines both errors — incorrectly doubled center coordinates (8,10) AND an incorrectly doubled (rather than quadrupled) right-hand side (18).

Solving with Desmos:

Step 1: Open Desmos and type circle M's equation: (x-4)^2+(y-5)^2=9

Step 2: Press Enter — Desmos graphs this circle; you can confirm its radius is 3 by inspecting the graph.

Step 3: Type circle P's equation using your calculated values: (x-4)^2+(y-5)^2=36

Step 4: Press Enter — Desmos graphs this larger circle. Visually confirm both circles share the SAME center point, and that circle P's radius (6) is exactly twice circle M's radius (3), confirming choice B.

Circles — Q8 [Hard] [Grid-In] (ID: 149021da)

Question ID: 149021da

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

A circle has center P, and points A and B lie on the circle. The measure of arc AB is $45°$ and the length of arc AB is 4π units. What is the length, in units, of the radius of the circle?

Correct Answer: 16 (Student-Produced Response — no answer choices given)

The Rule: The formula for arc length is: arc length = (central angle / 360°) × (2πr), where r is the radius. If you know the arc length and the central angle, you can solve this formula backwards to find the radius.

The Tip: Set up the arc length formula with the given angle and arc length, then solve for r algebraically — treat it just like any other equation with one unknown, isolating r on one side.

Why this is the answer: Using the arc length formula with angle 45° and arc length 4π: 4π = (45/360) × 2πr. Simplify 45/360 to 1/8: 4π = (1/8)(2πr) = πr/4. Multiply both sides by 4: 16π = πr. Divide both sides by π: r = 16.

Step-by-Step Solution:

Step 1: Recall the arc length formula: arc length = (central angle / 360°) × 2πr.

Step 2: Substitute the given values: 4π = (45/360) × 2πr.

Step 3: Simplify the fraction 45/360 by dividing both by 45: 45/360 = 1/8.

Step 4: The equation becomes: 4π = (1/8) × 2πr = (2πr)/8 = πr/4.

Step 5: Multiply both sides by 4 to clear the fraction: 16π = πr.

Step 6: Divide both sides by π: r = 16.

Step 7: This is a grid-in question — enter 16 as your final answer.

The Trap: A common error on this problem is forgetting to convert the angle fraction correctly (45/360 rather than accidentally using 45/180 or another incorrect fraction), which would throw off the entire calculation.

Solving with Desmos:

Step 1: Open Desmos and set up the equation directly: type 4*pi=(45/360)*2*pi*x

Step 2: Press Enter — since this is a one-variable equation, Desmos will solve for x and plot the solution.

Step 3: Look for the solution — it should show x = 16, confirming the radius and matching your grid-in answer.

Circles — Q9 [Hard] (ID: 10e059a6)

Question ID: 10e059a6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

The measure of angle A is $\frac{\pi}{216}$ radians. The measure of angle B is 36 times the measure of angle A. What is the value of $\cos B$?

Answer

A. 0

B. $\frac{1}{2}$

C. $\frac{\sqrt{2}}{2}$

D. $\frac{\sqrt{3}}{2}$

Correct Answer: D

The Rule: When one angle's measure is a MULTIPLE of another (like 'angle B is 36 times angle A'), first find the numeric value of the given angle, then multiply by that stated factor to find the second angle. Once you have the second angle's exact measure, apply the relevant trigonometric function directly.

The Tip: Keep angle measures in radians throughout the calculation if the problem starts in radians — converting to degrees and back introduces unnecessary opportunities for error. Trust the radian arithmetic, especially when working with fractions of π.

Why D is right: Angle A = π/216 radians. Angle B = 36 × angle A = 36 × (π/216) = 36π/216. Simplify this fraction: 36/216 = 1/6, so angle B = π/6 radians. The value of cos(π/6) is a standard reference angle value: √3/2.

Step-by-Step Solution:

Step 1: Identify the measure of angle A: π/216 radians (given).

Step 2: Calculate angle B using the given relationship: angle B = 36 × angle A = 36 × (π/216).

Step 3: Simplify this multiplication: 36 × π/216 = 36π/216.

Step 4: Reduce the fraction 36/216 by dividing both by 36: 36/216 = 1/6.

Step 5: So angle B = π/6 radians.

Step 6: Recall the standard value: cos(π/6) = √3/2 (this is one of the standard reference angles, equivalent to 30°).

Step 7: This matches choice D.

The Trap: Choice C (√2/2) is the most commonly picked wrong answer — it's the cosine value for a DIFFERENT standard angle (π/4, or 45°), confusing it with the correct angle π/6 (30°) found in this problem.

POE — Eliminate the other three:

✗ **A —** 0 would be the value of cos(π/2) (90°) — a completely different angle than the π/6 found in this problem.

✗ **B —** 1/2 is actually the value of sin(π/6), not cos(π/6) — a common mix-up between sine and cosine values for the same standard angle.

✗ **C —** √2/2 is the value of cos(π/4) (45°) — this belongs to a different standard angle than the π/6 calculated here.

Solving with Desmos:

Step 1: Open Desmos and calculate angle B first: type 36*(pi/216) — this confirms angle B = π/6 (approximately 0.524 radians).

Step 2: Calculate the cosine of this angle: type cos(36*(pi/216))

Step 3: Press Enter — Desmos will evaluate this to approximately 0.866, which matches √3/2 (you can verify by typing sqrt(3)/2 separately to compare the decimal values), confirming choice D.

Circles — Q10 [Hard] [Grid-In] (ID: 9f9112ab)

Question ID: 9f9112ab

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

The measure of angle L is 60 degrees. If the measure of angle L is $\frac{\pi}{n}$ radians, where n is a constant, what is the value of n?

Correct Answer: 3 (Student-Produced Response — no answer choices given)

The Rule: To convert an angle from DEGREES to RADIANS, multiply by the conversion factor (π/180°). A common shortcut for recognizable angles: 60° always equals π/3 radians — memorizing a few common conversions (30°=π/6, 45°=π/4, 60°=π/3, 90°=π/2) can save time.

The Tip: When you see a degree measure that's a 'nice' fraction of 180° (like 60°, which is 180°/3), you can often recognize the radian equivalent directly from memory rather than doing the full conversion calculation each time.

Why this is the answer: Converting 60° to radians: 60° × (π/180°) = 60π/180 = π/3 (simplifying the fraction 60/180 to 1/3). Since the problem states angle L's measure is π/n radians, and we've found it equals π/3, comparing directly: π/n = π/3, so n = 3.

Step-by-Step Solution:

Step 1: Convert the given angle measure from degrees to radians: 60° × (π/180°).

Step 2: Simplify the fraction 60/180 by dividing both by 60: 60/180 = 1/3.

Step 3: So 60° = π/3 radians.

Step 4: Compare this to the given form π/n: since angle L = π/3 = π/n, this means n = 3.

Step 5: This is a grid-in question — enter 3 as your final answer.

The Trap: A common error on this problem is miscalculating the fraction 60/180, perhaps simplifying incorrectly or forgetting to reduce it fully to 1/3, which would lead to an incorrect value of n.

Solving with Desmos:

Step 1: Open Desmos and convert 60 degrees to radians directly: type 60*pi/180

Step 2: Press Enter — Desmos will simplify this to π/3 (or show the decimal equivalent, approximately 1.047).

Step 3: Since the problem states the angle equals π/n, and you've found it equals π/3, this directly confirms n = 3 for your grid-in answer.

Circles — Q11 [Hard] [Grid-In] (ID: 27f1db42)

Question ID: 27f1db42

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

In the *xy*-plane, circle F is defined by the equation $(x+8)^2+(y+8)^2=25$. Circle G has the same center as circle F, and the radius of circle G is 2 units greater than the radius of circle F. Circle G is defined by the equation $(x+8)^2+(y+8)^2=p$, where p is a constant. What is the value of p?

Correct Answer: 49 (Student-Produced Response — no answer choices given)

The Rule: When two circles share the SAME CENTER (concentric circles) and one radius is described as a certain amount MORE than the other (an additive relationship, not a multiplicative one), first find the original radius, then ADD the stated amount to get the new radius — don't confuse this with problems where the new radius is a MULTIPLE of the original.

The Tip: Read the relationship between the two radii VERY carefully — 'is 2 units greater than' means ADD 2 to the radius (not to the radius squared, and not multiply by 2) — this additive relationship is different from 'twice as large' problems, which require multiplication instead.

Why this is the answer: Circle F has radius² = 25 (from the equation), so radius = 5. Circle G's radius is 2 units GREATER than circle F's radius: 5 + 2 = 7. Since circle G's equation uses p as the right-hand side (equivalent to radius²), p = 7² = 49.

Step-by-Step Solution:

Step 1: Identify circle F's radius from its equation $(x+8)^2 + (y+8)^2 = 25$: $radius^2 = 25$, so radius = $\sqrt{25} = 5$.

Step 2: Apply the given relationship: circle G's radius is 2 units GREATER than circle F's radius (an ADDITION, not a multiplication): circle G's radius = 5 + 2 = 7.

Step 3: Since circle G's equation is written as $(x+8)^2 + (y+8)^2 = p$, the value p represents circle G's radius SQUARED.

Step 4: Calculate p: $p = 7^2 = 49$.

Step 5: This is a grid-in question — enter 49 as your final answer.

The Trap: A common error on this problem is confusing this ADDITIVE relationship ('2 units greater') with a MULTIPLICATIVE one (like 'twice as large'), which would incorrectly lead to a radius of 10 (5×2) instead of the correct 7 (5+2), and consequently an incorrect p value of 100 instead of 49.

Solving with Desmos:

Step 1: Open Desmos and calculate circle F's radius first: type sqrt(25) — this confirms radius = 5.

Step 2: Calculate circle G's radius using the additive relationship: type 5+2 — this confirms radius = 7.

Step 3: Calculate p (radius squared): type 7^2

Step 4: Press Enter — Desmos will evaluate this to 49, confirming your grid-in answer.

Circles — Q12 [Hard] [Grid-In] (ID: 44d67c6c)

Question ID: 44d67c6c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

$$x^2 + y^2 - 10x - 8y - 14n = 0$$

The given equation represents circle A in the *xy*-plane, where n is a constant. Point $(11, 8)$ lies on circle B, which has the same center but twice the diameter as circle A. What is the value of n?

Correct Answer: -2 (Student-Produced Response — no answer choices given)

The Rule: To find the center and radius of a circle from an equation with x^2 and y^2 terms plus linear terms (not in standard form), use COMPLETING THE SQUARE on both the x-terms and y-terms. Once you have the standard form, you can relate the radius to any given point on a related (same-center) circle using the distance formula.

The Tip: When a problem involves TWO related circles (same center, different size) and gives you an actual POINT on the second circle, use the distance formula from the shared center to that point to find the second circle's radius directly — this is often more reliable than trying to track scale factors through multiple algebraic steps.

Why this is the answer: Complete the square on x^2-10x (giving $(x-5)^2-25$) and y^2-8y (giving $(y-4)^2-16$). The equation becomes $(x-5)^2 + (y-4)^2 = 25+16+14n = 41+14n$, so circle A has center (5,4) and radius² = 41+14n. Circle B has the same center (5,4) with a diameter twice circle A's — meaning circle B's radius is twice circle A's radius. Using the given point (11,8) on circle B, calculate circle B's actual radius via the distance formula: $\sqrt{[(11-5)^2+(8-4)^2]} = \sqrt{(36+16)} = \sqrt{52}$, so circle B's radius² = 52. Since circle B's radius = 2 × circle A's radius, circle B's radius² = 4 × circle A's radius²: $52 = 4\times(41+14n)$. Solving: $13 = 41+14n$, so $14n = -28$, giving $n = -2$.

Step-by-Step Solution:

Step 1: Complete the square for the x-terms in $x^2 - 10x$: take half of −10 (giving −5), square it (giving 25), rewrite as $(x-5)^2 - 25$.

Step 2: Complete the square for the y-terms in $y^2 - 8y$: take half of −8 (giving −4), square it (giving 16), rewrite as $(y-4)^2 - 16$.

Step 3: Substitute back into the original equation: $(x-5)^2 - 25 + (y-4)^2 - 16 - 14n = 0$.

Step 4: Rearrange to standard form: $(x-5)^2 + (y-4)^2 = 25 + 16 + 14n = 41 + 14n$. This means circle A has center (5,4) and radius² = 41+14n.

Step 5: Circle B shares the same center (5,4) but has TWICE the diameter of circle A — meaning circle B's radius is also twice circle A's radius.

Step 6: Use the given point (11,8) on circle B to find circle B's actual radius via the distance formula from the center (5,4): distance = $\sqrt{[(11-5)^2 + (8-4)^2]} = \sqrt{(36+16)} = \sqrt{52}$. So circle B's radius² = 52.

Step 7: Since circle B's radius = 2 × circle A's radius, squaring both sides: circle B's radius² = 4 × circle A's radius². Substitute: $52 = 4 \times (41+14n)$.

Step 8: Divide both sides by 4: $13 = 41 + 14n$.

Step 9: Solve for n: $14n = 13 - 41 = -28$, so $n = -28/14 = -2$.

Step 10: This is a grid-in question — enter −2 as your final answer.

The Trap: A common error on this problem is forgetting that DOUBLING the diameter means the radius SQUARED (used in the equation) actually QUADRUPLES, not just doubles — using a factor of 2 instead of 4 when relating circle A's and circle B's radius-squared values would lead to an incorrect value of n.

Solving with Desmos:

Step 1: Open Desmos and complete the square verification: type the original equation x^2+y^2-10x-8y-14*n=0 with n as a slider — Desmos will offer to create a slider.

Step 2: Calculate circle B's radius using the given point and center: type sqrt((11-5)^2+(8-4)^2) — this confirms circle B's radius is $\sqrt{52}$.

Step 3: Set up the relationship: circle B's radius² (52) should equal 4 times circle A's radius² (41+14n). Type: 52=4*(41+14x) and solve for x using Desmos (treating it as an equation to solve).

Step 4: Desmos will show the solution $x = -2$, confirming $n = -2$ for your grid-in answer.